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GIHON Web based E-commerce

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Approval sheet

We, the undersigned, members of the Board of Examiners of the final open defense by List of students listed above, have read and evaluated their project entitled "**GIHON Web Based E-commerce**" and examined the candidates. This is, therefore, to certify that the project has been accepted in partial fulfillment two-month ago in computer science department

Examining committee members

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ABSTRACT

In today's fast-changing business environment, it's extremely important to be able to respond to client needs in the most effective and timely manner. If your customers wish to see your business online and have instant access to your products or services.

Online Shopping is a lifestyle e-commerce web application, which retails various fashion and lifestyle products. This project allows viewing various products available and enables registered users to purchase desired products instantly using the Chappa payment processor (Instant Pay) .

In order to develop an e-commerce website, a number of technologies must be studied and understood. These include multi-tiered architecture, server and client side scripting techniques, implementation technologies such as web HTML,CSS, programming language (such as java script and nodejs) and relational databases such as MySQL. This is a project with the objective to develop a basic website where a consumer is provided with a shopping cart application and also to know about the technologies used to develop such an application.

Acknowledgement

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CHAPTER ONE

1. INTRODUCTION

The project is an “GIHON e-commerce” web application which provides a platform for buying and selling of e-commerce products such as books, electronic items, shoes etc. As part of the e-commerce platform, users can sign up and log-in into the application using a general email-id. Once logged in, users will then be able to add products to their cart and then checkout for final shipping. Users can also provide reviews to the products which they bought. Users can search products based on categories and are made aware of products with special discount offers.

Our primary objective is to develop an outstanding online marketplace that caters to four distinct user groups, each with unique roles and experiences:

- Admin: The admin oversees and manages the entire system, ensuring smooth operation and maintenance of the platform.

- Clients: These are the everyday shoppers who seek a convenient, personalized shopping experience. Our platform is designed to meet their needs by providing ease of access and tailored product recommendations.
- Merchants: The platform offers a dedicated space for sellers of all kinds, ranging from small startups to well-established brands. Merchants can showcase and sell their products in a user-friendly environment.

Streamline Buying and Selling Processes Our user-friendly platform will empower both clients and merchants with intuitive tools for seamless transactions. Clients will enjoy a frictionless shopping experience with clear product navigation, advanced search options, and secure payment gateways. Merchants will benefit from streamlined product management, efficient order processing, and built-in marketing tools to boost sales.

Improve Communication and Collaboration: An integrated communication system will foster seamless interaction between all user groups. Real-time notifications and alerts will keep everyone informed on order status, inquiries, and support requests.

1.1. Background of the project

This proposal tackles that by creating a revolutionary, inclusive online marketplace. Imagine a platform designed for three key roles: clients, merchants (businesses of all sizes) and (individuals connecting buyers and sellers). We go beyond just buying and selling. Our platform streamlines the entire process for everyone – user- friendly navigation for shoppers, effortless store setup for businesses, and dedicated support for helpers. An integrated communication system and real-time updates ensure smooth collaboration. Additionally, powerful analytics empower everyone with valuable insights. This proposal lays the groundwork where everyone involved gets something they want or benefits in some way, encouraging an online environment where everyone involved is successful and active for all participants.

1.2. Statement of the Problem

Ethiopia's ecommerce sector is still in its early stages, with several challenges hindering the growth of online shopping. These challenges include a lack of access to modern payment systems, limited internet penetration in rural areas, low trust in online transactions, and insufficient infrastructure to support large-scale ecommerce. Many local businesses struggle to transition from traditional brick-and-mortar stores to an online marketplace, missing out on the potential to reach a broader audience.

Additionally, consumers face difficulty accessing a wide range of products conveniently, while intermediaries, such as s or logistics providers, are not integrated into existing online platforms. As a result, there is a clear gap in the market for a comprehensive, localized eCommerce solution that addresses these challenges and supports all stakeholders, including merchants, customers, and service providers.

1.3. Objective of the project

1.3.1. General Objective

The main objective of this project is to build an efficient and user-friendly ecommerce platform tailored to the Ethiopian market. The platform will aim to bridge the gap between merchants, clients, and intermediaries by providing a secure, reliable, and scalable solution that enables online transactions, product showcasing, and service facilitation in a convenient way.

1.3.2. Specific Objectives

- Perform Requirement Analysis conduct a thorough analysis to identify both the functional and non-functional requirements of the system, ensuring the platform meets the needs of all stakeholders, including clients, merchants, s, and administrators
- Identify Problems in the Existing System Investigate the limitations and challenges of the current ecommerce practices in Ethiopia to design a solution that addresses these issues effectively, such as improving payment systems, and user experience.
- Manage Records Efficiently Develop a robust system to manage and track essential records, including customer profiles, product details, transaction histories, and employee data, to ensure smooth operation and efficient record-keeping across the platform.

1.4. Feasibility Analysis

1.4.1. Operational Feasibility

- The proposed ecommerce web application is operationally feasible, as it provides a user-friendly platform that simplifies buying and selling processes for clients, merchants, and s. Clients can easily search for products, add them to their carts, and securely check out, while merchants benefit from streamlined product management and order processing. Admins will maintain the system, manage users, and ensure smooth operations, overseeing critical aspects like promotions and product reviews. Additionally, the platform's scalability and built-in communication tools will enable seamless interactions between all user groups, supporting real-time notifications and updates.
- The application will also integrate secure payment gateways. Merchants will have tools to manage discounts and promotions, while clients can provide feedback through product reviews. Routine maintenance and system updates will be handled by the admin, ensuring the platform runs efficiently as it grows. Overall, the operational feasibility of this ecommerce platform is strong, meeting the needs of the Ethiopian market by creating a reliable and accessible online marketplace.

1.4.2. Technical Feasibility

- Technology Stack: Identify the programming languages, frameworks, and databases needed to develop the platform. Consider factors like scalability, security, and ease of integration with payment gateways like Tele birr.
- Development Team: Evaluate the resources required to build and maintain the platform. Consider the skills and experience necessary for the successful implementation of the project.
- Technical Challenges: Address any potential technical hurdles. This could involve integrating various functionalities, ensuring secure data storage.

1.4.3. Economic Feasibility

The proposed ecommerce platform is economically feasible as it addresses the growing demand for online shopping in Ethiopia. By connecting merchants with clients and ensuring product quality through oversight, the platform mitigates risks for buyers while providing a reliable marketplace for sellers. The investment in development and operations is expected to be justified by the revenue generated from merchant fees, commissions, and transaction-based earnings, making it a sustainable venture that supports job creation and expands market access for merchants.

1.4.3.1. Cost-Benefit Analysis

The cost-benefit analysis identifies various costs associated with the project, including platform development, infrastructure, logistics integration, marketing, and ongoing maintenance. These costs are weighed against potential benefits such as revenue from merchant fees, client transactions, and commissions. Merchants gain access to a broader customer base, while clients enjoy a secure shopping experience with verified products. Overall, the expected benefits will provide a favorable return on investment, contributing to the platform's success and growth.

1.4.3.2 Cost of the Project

The overall cost of the project includes various components such as initial development expenses for creating the platform, database setup, and logistics integration, along with ongoing costs for infrastructure, marketing, and maintenance. This comprehensive investment is crucial for ensuring the operational efficiency and sustainability of the platform, supporting its long-term growth and the economic ecosystem surrounding it.

1.4.4 .Schedule Feasibility

The schedule feasibility of the proposed ecommerce platform is structured into two main phases: documentation and implementation. The documentation phase will span 2 months, during which requirements gathering, design specifications, and planning will be finalized. Following this, the implementation phase will extend for up to 4 months, encompassing development, testing, and deployment of the platform. This timeline allows for a comprehensive approach to building the platform, ensuring that all necessary features are developed, rigorously tested, and effectively launched. By allocating sufficient time for both documentation and implementation, the project remains on track for a successful rollout, addressing the needs of all stakeholders involved.

1.5. Scope and limitation of the project

1.5.1. Scope of the Project

The project scope will encompass the following:

- Development of the e-commerce platform with functionalities for clients, merchants, and admin.
- Integration of secure payment gateways.
- Development of a communication system for messaging and notifications.
- Implementation of analytics and reporting tools.
- Design of a user-friendly interface for all user types.

1.5.2. Limitation of the project

- **Technical Challenges**: Developing a robust and secure platform may require significant technical expertise and resources, which can lead to delays and increased costs.
- **Security Concerns**: Protecting sensitive customer data, such as payment information, is critical. Breaches can damage trust and lead to financial losses.
- **Dependence on Technology**: ecommerce relies heavily on internet connectivity and technology. Any downtime or technical issues can disrupt sales and customer satisfaction.
- **Competition**: The online marketplace is highly competitive, making it challenging for new entrants to establish a foothold against established brands.
- **Logistics and Fulfillment**: Managing inventory, shipping, and returns can be complex and costly, particularly for businesses operating on a global scale.
- **Customer Trust**: Building trust with customers can take time, especially for new ecommerce platforms. Concerns about product quality and return policies may deter purchases.

- **Regulatory Compliance**: Businesses must navigate various legal and regulatory requirements, such as data protection laws and tax regulations, which can vary by region.
- **Market Saturation**: Some niches may be oversaturated, making it difficult to differentiate products and attract customers.
- **User Experience**: A poor website design or navigation can lead to high bounce rates and lost sales, necessitating ongoing investments in user experience improvements.
- **Changing Consumer Behavior**: Rapid shifts in consumer preferences and shopping habits require ecommerce businesses to adapt quickly, which can be challenging.
- Supports only English language

1.6. Methodology

The system development methodology we selected for our system is the waterfall model, because water fall model is a linear and sequential approach to software development. It consists of distinct phases where each phase must be completed before the next one begins.

The main phases are

- Requirement analysis
- System analysis
- System Design
- Implementation
- Testing
- Maintenance

1.6.1. Data collection method

To get precise data, the team member will use the following data collection techniques. Those are:

Interview: Direct interviews are a qualitative data collection method that involves face-to-face, phone, or video interactions with customers or stakeholders. This method can provide deep insights into customer behavior, preferences, and experiences..

Surveys and Questionnaires:

- **Customer Feedback:** Use online surveys to gather insights on customer satisfaction, preferences, and suggestions.
- **Post-Purchase Surveys:** Collect feedback after transactions to understand the shopping experience.

Direct observation: Direct observation is a qualitative data collection method where researchers watch and record customer behavior in real-time, either in physical stores or through online platforms. This method provides valuable insights into how customers interact with products and services.

1.6.2. tools will use in the system

Table 1: tools will use in the system

Tool category	Tool name	Description
Server-side programming	Nodejs without frame work	A widely-used server-side scripting language for web Development.

Client-side Programming	HTML	A standard markup language for creating the structure of web pages.
	CSS	A style sheet language used to describe the presentation of a document written in a markup language like HTML.
	JAVASCRIPT	A high-level, dynamic, and interpreted programming Language for creating interactive web pages.
Database management	MYSQL	A popular open-source relational database management system.
Code editor	Visual Studio code	A free and open-source code editor with support for Various programming languages.
Documentation	Google doc	A word processing software for creating and editing documents.
Presentation	Microsoft office power point 2016	A presentation software for creating and delivering Slide-based presentations.
Diagramming	EdrawMax,Enterprise architecture,Draw.io	A comprehensive diagramming software for creating various types of diagrams, such as use case diagrams, activity diagrams, sequence diagrams, and class

		Diagrams.
Web Server	XAMPP	A free and open-source cross-platform web server solution stack package, which includes Apache, MySQL, Nodejs

1.7. Significance of the project

- Online Product Catalog and Listing
- Payment Integration
- Create an Online Marketplace
- User Registration and Authentication
- Security and Data Protection
- Marketing and Promotion

1.7.1. Significance for Clients

- Provides a user-friendly interface with advanced search and filter options for easy product discovery and a smooth purchase process

- Secure payment gateways and a ratings and reviews system for informed purchasing decisions
- Save time, man power and cost
- To avoid car Rescue
- Shopping Cart and Checkout
- Search item and Filtering

1.7.2. Significance for Merchants

- Simplified Online Presence: Enables setting up online stores with an easy onboarding process.
- Efficient Sales Management: Provides tools for product management, order processing, fulfillment, and tracking.
- Expanded Market Reach: Integrates and network to expand reach and target new customers.
- Enhanced Sales Potential: Offers marketing and promotional tools to increase product visibility and sales.
- Data-Driven Decisions: Provides access to analytics dashboards for tracking sales performance and making informed business decisions.

1.7.3. Significance for s

- Income Opportunity: Creates a platform for individuals or companies to generate income through sales commissions.
- Centralized Management: Offers a user-friendly dashboard to track sales performance and manage customer interactions.
- Product Knowledge: Provides access to training and educational resources to enhance product knowledge and sales skills.
- Improved Communication: Facilitates communication with clients through integrated messaging tools

1.8. Budget and time schedule of the system

1.8.1. Budget of the system

Estimated assume budget for the e-commerce system project, including tools and papers

Table 2: Budget for tools and software

Budget item	Estimated Cost
Tools and software	Open source
Desktop, Laptop	50,000
Integrated Development Environment (IDE)	Open source
Version Control System (e.g., Git)	Open source
Database Management System	10,000
Web Server and Application Server	10,000
Payment Gateway Integration	Open source
Total	

Table 3: Budget for paper

Budget item	Estimated cost
Papers and Documentation	2,000 ETB
System architecture and Design Documents	1,500 ETB
User stories and Requirements specification	1,000 ETB
Test plans and test cases	1,000 ETB
Operation and maintenance manual	2,000 ETB
Total	

1.8.2. Time schedule of the project

brief overview of the typical time schedule for an e-commerce project

Table 4: Time schedule

program	Time line	description
Planning and Design	2-4 weeks	Define project requirements, conduct market research, and design the user experience.
Development	4-6 weeks	Set up the e-commerce platform and build the website, integrating with payment and other services.
Content Creation	2-4 weeks	Create product descriptions, images, and other website content.

Deployment and Launch	1-2 weeks	Finalize hosting, testing, and staff training before rolling out the live website.
Ongoing Maintenance	continuous	Monitor performance, update content, and optimize the site over time.

1.9. Communication plan

The communication plan for the ecommerce platform project will establish clear channels and protocols to ensure effective information flow among stakeholders, including merchants, clients, and the admin team. Regular updates will be provided through weekly progress meetings, supported by email summaries that outline key developments and action items. A dedicated project management tool will facilitate ongoing collaboration, allowing team members to share documents, track tasks, and address any issues in real-time.

Additionally, stakeholders will have access to a centralized communication platform for immediate inquiries and support, fostering a collaborative environment that enhances engagement and keeps all parties informed throughout the project's lifecycle.

1.10 .Team composition of the project

Table 5: tasks of group member

Task no.	Task list	Accomplished by			
		Abyissinia	kelem	Temessgen	Alemenew

1	Project organization and progress controlling	✓	✓	✓	✓
2	Programming activist interface building and mapping	✓	✓	✓	✓
3	Database designing			✓	✓
4	content development	✓	✓		
5	Interface selector, UI analyzing	✓	✓	✓	✓
6	Graphic designing			✓	✓

7	Coding	✓	✓	✓	✓
8	Document analyzing			✓	v
9	Communicating and decision making	✓	✓	✓	✓
1	Presenting and giving orientation to advisor and department staff member	✓	✓	✓	✓
1	preparing documentation	✓	✓	✓	✓
1	Gathering requirement	✓	✓	✓	✓

1	Analyzing and specifying requirement	✓	✓	✓	✓
1	Receiving feedbacks	✓	✓	✓	✓

CHAPTER TWO SYSTEM REQUIREMENT SPECIFICATION

2. INTRODUCTION TO THE EXISTING SYSTEM

The current ecommerce system in Ethiopia is fragmented and primarily operates through informal channels, which significantly limits accessibility and efficiency for both merchants and clients. Many businesses lack an online presence, making it difficult for them to reach a broader customer base. Clients often face challenges in finding quality products, as there is no centralized platform for secure and convenient online shopping. Additionally, issues such as product quality assurance and reliable payment processing lead to mistrust in online transactions, resulting in a suboptimal shopping experience. This situation underscores the urgent need for a comprehensive ecommerce platform that can integrate various elements to improve the overall shopping experience and support local businesses.

- Fragmented Marketplace: Reliance on informal channels limits efficiency and accessibility.
- Lack of Online Presence: Many merchants struggle to reach a wider audience without an online platform.
- Quality Assurance Issues: Absence of a centralized system makes it hard for clients to find verified products.
- Trust and Security Concerns: Mistrust in online transactions due to payment processing issues.
- Ineffective Communication: Lack of communication channels complicates the buying and selling processes.
- Need for Comprehensive Solution: Urgent need for an integrated platform to enhance shopping experiences and support local businesses.

2.1. Players in the Existing System

The existing ecommerce landscape in Ethiopia involves several key players, each contributing to the overall dynamics of buying and selling:

- Merchants: Local businesses, including small startups and established brands, who sell products but often lack an effective online presence to reach customers.
- Clients: Everyday shoppers seeking a convenient and reliable shopping experience but facing challenges in finding quality products and navigating informal purchasing channels.
- Payment Providers: Banks and financial institutions that handle transactions, providing payment processing solutions but may face issues with security and trust among users.
- Logistics Providers: Local delivery services responsible for transporting goods, which can impact the efficiency and reliability of product delivery.

- Regulatory Bodies: Government agencies that oversee business operations and ecommerce regulations, influencing the overall framework within which these players operate.

2.2. Major Functions/Activities in the Existing System

The existing ecommerce system can be broken down into key functions/activities characterized by inputs, processes, and outputs:

Major Functions/Activities in the Existing System

1. Product Listing
    - Inputs: Product details (name, description, price, images) and Merchant information
    - Processes: Merchants enter product information through informal channels (e.g., social media, phone)
    - Outputs: Listed products available for clients to view

2. Customer Browsing

- Inputs: Client search queries and Product categories
    - Processes: Clients search for products using keywords or categories in informal platforms
    - Outputs: Display of relevant products for clients

3. Order Placement
    - Inputs: Selected products and client details (name, contact information)
    - Processes: Clients communicate their orders to merchants via phone calls or messages
    - Outputs: Order details confirmed by both client and merchant

4. Payment Processing
    - Inputs: Payment information (cash, bank transfer details)
    - Processes: Clients make payments through agreed methods, often involving direct transactions
    - Outputs: Transaction confirmation and receipt (if applicable)

5. Quality Assurance

- Inputs: Product samples
- Processes: s inspect products to verify quality before delivery
- Outputs: Verified products approved for delivery

6. Delivery Coordination

- Inputs: Order details and delivery addresses
- Processes: Merchants or s arrange for local logistics to deliver products
- Outputs: Delivered products to clients, including delivery confirmations

7. Customer Feedback Collection

- Inputs: Client reviews and ratings
- Processes: Clients provide feedback after receiving products, often through informal channels
- Outputs: Customer reviews that are shared with potential buyers and merchants

8. Communication

- Inputs: Inquiries from clients or merchants
- Processes: Clients and merchants communicate via phone, messages, or in-person meetings
- Outputs: Resolved inquiries and improved understanding between parties

2.3. Business Rules for the Existing E-commerce System

The following business rules outline the foundational guidelines that govern the operations and interactions within the existing ecommerce system in Ethiopia:

- Merchant Registration: Only registered merchants are allowed to list and sell products on the platform. Registration requires verification of business credentials.

- Product Quality Standards: All products listed must meet specified quality standards, and merchants are responsible for ensuring their products are accurate and of high quality.
- Pricing Transparency: Merchants must clearly display prices for all products, including any applicable taxes or fees, ensuring clients are fully informed before making a purchase.
- Order Confirmation: Once a client places an order, an order confirmation must be sent to both the client and the merchant, detailing the order specifics.
- Payment Processing: All transactions must be conducted through approved payment methods to ensure security and traceability, minimizing the use of cash transactions.
- Compliance with Regulations: All activities conducted through the platform must comply with local laws and regulations governing ecommerce and consumer rights.
- Data Privacy: Client and merchant data must be protected, ensuring compliance with data protection regulations and maintaining customer trust.

2.4 Reports Generated in the Existing System

1. Sales Reports
    - Summarizes total sales, revenue by product category, and merchant performance.
    - Helps merchants track sales trends and adjust inventory and pricing strategies.
2. Inventory Reports
    - Monitors stock levels, identifies low-stock items, and highlights fast-moving products.
    - Assists merchants in managing inventory effectively and ensuring product availability.
3. Order Reports
    - Provides details on orders, including their status (pending, completed, or canceled).
    - Helps both merchants and s track order progress and resolve issues promptly.
4. Customer Feedback Reports
    - Compiles customer ratings and reviews for products and services.

- Enables merchants to enhance product quality and customer satisfaction.

5. Financial Reports

- Tracks overall financial performance, including payment transactions, profits, and expenses.
- Assists merchants and administrators in monitoring cash flow and profitability.

2.5. Proposed Solution for the new ecommerce System

The proposed solution is to develop a comprehensive ecommerce platform that enhances the existing system by addressing its limitations and improving efficiency for all user groups' admins, clients, and merchants. The new system will focus on providing a seamless experience for both buyers and sellers, incorporating the following features:

- Centralized Product Management: A streamlined interface for merchants to easily list, update, and manage their products with real-time inventory tracking.
- User-Friendly Shopping Experience: Enhanced search, filtering, and browsing features for clients to easily find products, supported by personalized recommendations and a secure checkout process.
- Robust Payment: Secure online payment options with real-time tracking for clients and merchants.
- Admin Dashboard and Analytics: A powerful admin interface that oversees the entire system, provides analytics on sales, customer behavior, inventory levels, and ensures compliance with regulations.

2.6. Definitions and Acronyms

2.6.1. Definitions

To facilitate a clear understanding of the document, the following terms, acronyms, and abbreviations are defined:

- **SRS**: System Requirement Specification

- **E-commerce**: Electronic commerce; the buying and selling of goods and services over the internet.
- **UI**: User Interface; the means by which a user interacts with the application.
- **UX**: User Experience; the overall experience a user has when interacting with the application.
- **API**: Application Programming Interface; a set of protocols for building and interacting with software applications.
- **Payment Gateway**: A service that authorizes credit card or direct payments for e-commerce transactions.
- **CMS**: Content Management System; software that helps create, manage, and modify content on a website

2.6.2. Abbreviations

Table 6: abbreviation table

MB	Mega byte
RAM	Random access memory
GB	Giga byte
PC	Personal computer
SRS	System requirement specification
E-commerce	Electronic commerce
UI	User interface
CMS	Content management system
API	Application programming interface
HTML	Hypertext markup language

PHP	Hypertext preprocessor
CSS	Cascading style sheet
JS	JavaScript
HTTP	Hypertext transfer protocol
MYSQL	Structured query language
WCAG	Web Content Accessibility Guidelines
ETB	Ethiopian birr
RESTful	Representational State Transfer
CRM	Customer relationship management
IOS	IPhone operating system
PCI DSS	Payment Card Industry Data Security Standard
I Product	Interface product
I order	Interface order
I payment	Interface payment
I customer	Interface customer
I shopping cart	Interface shopping cart
I delivery	Interface delivery
chatAI	

2.7. Specific Requirements

This section details the functional and non-functional requirements for the web-based e-commerce application. Functional requirements specify what the system should do, while non-functional requirements define how the system should perform those functions.

2.7.1. Functional Requirements

2.7.1.1. User Registration and Authentication

- **Registration Process**: Users must be able to create an account by providing essential information, such as name, email address, and password. An email verification step should be included to confirm the registration.
- **Login/Logout**: Users should be able to log in using their registered email and password, with options for password recovery and "remember me" functionality.
- **Role Management**: The system must differentiate between customer, and administrator roles, providing appropriate access and functionalities based on the user type.

2.7.1.2. Product Catalog Management

- **Product Listings**: merchants must be able to add, edit, and remove products from the catalog, including product details such as name, description, price, images, and stock levels.
- **Categories and Tags**: The system should allow categorization of products for easy navigation and searching. Vendors can assign tags to products for better discoverability.

2.7.1.3. Shopping Cart and Checkout Process

- **Cart Functionality**: Users should be able to add products to a shopping cart, view cart contents, and modify item quantities or remove items as needed.
- **Persistent Cart**: The system must retain the user's cart contents between sessions, allowing for a seamless shopping experience.
- **Checkout Process**: The checkout process must be streamlined, guiding users through entering shipping information, selecting shipping methods, and reviewing orders before finalizing purchases.

2.7.1.4 Payment Processing

- **Payment Gateway Integration**: The application must support payment methods like chappa .
- **Order Confirmation**: Upon successful payment, users should receive an immediate confirmation email with order details and tracking information.

2.7.1.5 Order Management

- **Order Tracking**: Users should be able to track the status of their orders through the application, receiving notifications for order updates (e.g., shipped, delivered).
- **Order History**: The system must allow users to view their past orders and re-order items easily.

2.7.1.6 Search Functionality

- **Search Bar**: The application must include a prominent search bar that allows users to search for products by name, category, or tags.
- **Filters and Sorting**: Users should be able to filter search results by price,

Availability, and other criteria. Sorting options should include relevance, price (low to high and vice versa), and newest arrivals.

- **Auto-suggestions**: As users type in the search bar, the system should provide auto-suggestions for products and categories.

2.7.1.7 Reporting and Analytics

- **Sales Reports**: admin must have access to detailed sales reports, including total sales, average order value, and sales trends over time.
- **User Analytics**: The system should track user behavior, such as product views, purchase history, and abandoned carts, to generate insights for improving marketing strategies.
- **Dashboard**: A user-friendly dashboard should be provided for administrators, summarizing key metrics and performance indicators.

2.7.2. Non-Functional Requirements

2.7.2.1 *Performance Requirements*

- **Response Time**: The application should have a response time of less than 2 seconds for user interactions, including page loads, searches, and checkout processes.
- **Load Handling**: The system must support at least 10,000 concurrent users without degradation in performance, ensuring a smooth user experience during peak times.

2.7.2.2 Security Requirements

- **Data Protection**: All sensitive user data, including passwords and payment information, must be securely encrypted both in transit and at rest.
- **User Authentication**: Implement multi-factor authentication (MFA) to enhance security for user accounts, especially for vendors and administrators.
- **Vulnerability Management**: Regular security assessments and penetration testing must be conducted to identify and mitigate potential vulnerabilities.

2.7.2.3 Usability Requirements

- **User Interface Design**: The application must feature a clean, intuitive design that enhances user experience, ensuring easy navigation and accessibility.
- **Mobile Responsiveness**: The application should be fully responsive, providing an optimal viewing experience on various devices and screen sizes, including smartphones and tablets.
- **Onboarding Process**: First-time users should be guided through an onboarding process that highlights key features and functionalities.

2.7.2.4 Reliability and Availability

- **Uptime Guarantee**: The application must maintain an uptime of 99.9%, ensuring continuous access for users.
- **Backup and Recovery**: Regular backups should be implemented to ensure data integrity, with a disaster recovery plan in place to restore operations in case of system failure.

2.7.2.5 Scalability

- **Horizontal Scalability**: The system must be designed to easily add additional servers or resources as user demand increases, ensuring consistent performance.
- **Modular Architecture**: The application should employ a modular architecture, allowing for the addition of new features and integrations without disrupting existing functionalities.

2.7.3 External Interface Requirements

This section outlines the external interface requirements for the web-based e-commerce application. These interfaces are critical for ensuring that the application interacts effectively with users, hardware, software, and communication systems, facilitating a seamless experience for all stakeholders.

2.7.3.1. User Interfaces

The user interface (UI) is a crucial aspect of the e-commerce application, as it determines how users interact with the system. The UI must be intuitive, responsive, and accessible across various devices. The following elements are essential:

2.7.3.2. Design Guidelines

- **Consistency**: All UI elements should maintain a consistent design language, including color schemes, typography, and button styles, to enhance usability and brand recognition.
- **Accessibility**: The interface must comply with Web Content Accessibility Guidelines (WCAG), ensuring that users with disabilities can navigate the application. This includes providing text alternatives for images, keyboard navigation, and adjustable text sizes.
- **Responsive Design**: The UI must adapt to different screen sizes and orientations, ensuring an optimal experience on desktops, tablets, and mobile devices. Media queries should be used to adjust layouts and components dynamically.

2.7.3.3. Key UI Components

- **Navigation Bar**: A clear and concise navigation bar should be present on all pages, providing easy access to product categories, user account settings, and support resources.
- **Product Pages**: Each product page should display high-quality images, detailed descriptions, pricing, and user reviews. A prominent “Add to Cart” button should be easily accessible.
- **Shopping Cart**: The shopping cart interface should allow users to view, modify, and remove items, displaying the total cost and any applicable discounts or shipping fees.
- **Checkout Flow**: The checkout process must be streamlined, with a multi-step form guiding users through entering shipping information, selecting payment methods, and reviewing orders.

2.7.3.4 User Feedback Mechanisms

- **Error Messages**: Clear and informative error messages should be displayed when users encounter issues, such as invalid input or failed transactions. Messages should guide users on how to rectify the problem.
- **Confirmation Notifications**: Users should receive confirmation messages upon completing actions, such as successful registration, adding items to the cart, and completing purchases.

2.7.4 Hardware Interfaces

The e-commerce application operates primarily as a web-based platform; however, it may interact with various hardware components, particularly during payment processing and order fulfillment. Key hardware interfaces include:

- The application demands that all the PCs must be present on the internet.
- The PC should be sufficiently fast with adequate memory at least 64 MB RAM and 2 GB hard –disk space is required to run this application.
- Screen resolution of at least 800*600 required to properly view the screen.

• Computer systems will be in the networked environment as it is a multi-user system.

2.7.4.1. Mobile Devices

- **Smartphones and Tablets**: The application must be optimized for touch interfaces on mobile devices, ensuring that buttons and links are adequately sized for easy tapping.
- **Location Services**: If applicable, the application may utilize GPS functionality to provide location-based services, such as finding nearby stores or estimating shipping times.

2.7.5. Software Interfaces

The communication between client and server is asynchronous. This will help to handle a large number of users simultaneously. The portal should support all major web browsers that will make it convenient for the user to access our system with ease. The back- end i.e., the database services will be used to a great extent and hence it will be quiet efficiently designed.

Hosting Service:

Source: http://10.127.0.11/

This hosting service will serve as the hosting of the MySQL relational database and the web site Javascripts and HTML web pages. The hosting service will have the necessary software requirements to run MySQL and code.

MySQL Database:

Specification Number: 5.0

Source: http://www.mysql.com

This relational database will serve as the backbone of the **web Based E-commerce system**. It will allow the users information to be captured, stored and then displayed in various forms to each user who visits the web site. The MySQL module will be utilized through an object-oriented approach in order to successfully access and modify database tables using MySQL.

NODE.JS with Express js Framework

Specification Number: 20.x

Source: https://nodejs.org

The Node.js runtime environment will interface and interact with the MySQL database to retrieve and process data, providing dynamic content to users in the form of HTML documents rendered in a compatible web browser. A modular and object-oriented programming approach will be adopted, utilizing JavaScript and modern frameworks (such as Express.js), to ensure high performance, scalability, code portability, reusability, and long-term maintainability of the system.

2.7.6 Communications Interfaces

Effective communication interfaces are essential for ensuring that the e-commerce application operates smoothly and provides timely information to users. Key communication interfaces include:

2.7.6.1. Email Services

- Transactional Emails: The system should utilize email services to send transactional emails, including registration confirmations, order confirmations, shipping notifications, and password reset links.
- Promotional Emails: Integration with email marketing platforms to manage and send promotional emails, newsletters, and user-specific offers.

2.7.6.2 Push Notifications

- Mobile Notifications: If a mobile application is developed, the system should support push notifications to inform users about order updates, special promotions, and personalized recommendations.
- Web Notifications: For web users, browser notifications can be used to alert users about cart reminders, new product launches, and upcoming sales.

2.7.6.3 Client Support Interfaces

- Live Chat: Integration with live chat software to provide real-time assistance to users, enabling them to ask questions and receive support during their shopping experience.
- Helpdesk Software: The application should integrate with helpdesk systems to manage user inquiries and support tickets, ensuring timely responses and resolution of issues.

2.8 System Features

This section outlines the key features of the web-based e-commerce application. Each feature is designed to enhance user experience, improve operational efficiency, and provide valuable tools for both customers and vendors.

2.8.1 User Account Management

User account management is a critical feature that enables users to create, manage, and secure their accounts within the e-commerce platform. This feature includes several key components:

2.8.1.1 User Registration

- Account Creation: Users can register for an account by providing essential information such as name, email address, and password. The registration process should include email verification to ensure the authenticity of user accounts.

2.8.1.2 Profile Management

- Profile Editing: Users can view and edit their profiles, including personal information, shipping addresses, and payment methods.
- Password Management: Users can update their passwords and utilize password recovery options, such as sending a reset link to their registered email.

2.8.1.3 Role-Based Access Control

- User Roles: The system supports multiple user roles, including customers, and administrators. Each role has specific permissions and access to features tailored to their needs.

- Admin Controls: Administrators can manage user accounts, including approving or suspending accounts, resetting passwords, and monitoring user activity.

2.8.1.4 Security Features

- Authentication (2FA): Users can enable 2FA for added security, requiring a second form of verification in addition to their password during login.
- Data Protection: All user data, including personal information and payment details, are encrypted and stored securely in compliance with data protection regulations.

2.8.2. Management

Efficient management is essential for admin to maintain stock levels and streamline operations. This feature provides tools for managing product inventory effectively:

2.8.2.1 Product Listing and Categorization

- Add/Edit Products: Vendors can easily add new products to the inventory, including details such as name, description, price, images, and stock levels. They can also edit existing product listings.
- Categories and Tags: Products can be organized into categories and tagged for better navigation and search ability, enhancing the user experience.

2.8.2.2 Reporting and Analytics

- Sales Analytics: Vendors can access reports on product performance, including sales trends, popular items, and seasonal variations. This data helps in making informed inventory decisions.
- Forecasting Tools: The system can provide inventory forecasting tools based on historical sales data, helping vendors anticipate demand and optimize stock levels.

2.8.3. Client Support

Providing excellent customer support is crucial for enhancing user satisfaction and loyalty. This feature offers various tools to assist users effectively:

2.8.3.1. Help Center

- Knowledge Base: A comprehensive help center should be available, containing articles, FAQs, and guides to assist users in resolving common issues independently.
- Search Functionality: Users can search the knowledge base to find relevant articles quickly, helping them access information efficiently.

2.8.3.2. Live Chat Support

- Real-Time Assistance: Users can engage with customer support representatives through live chat for immediate assistance with their inquiries or issues.
- Chat History: The system should maintain a history of chat interactions, allowing users to refer back to previous conversations if needed.

2.8.3.3. Support Ticket System

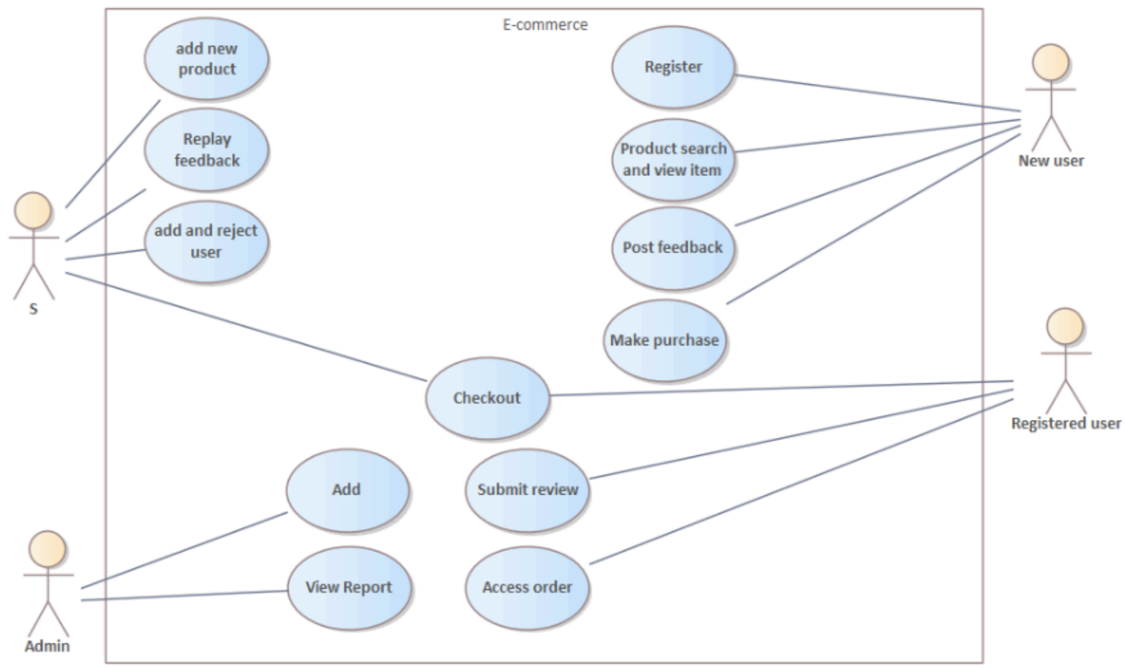
- Ticket Submission: Users can submit support tickets for issues that require more in-depth assistance. Each ticket should be assigned a unique ID for tracking.
- Status Tracking: Users can track the status of their support tickets, receiving notifications when their ticket is updated or resolved.

2.8.3.4. Feedback Mechanism

- User Feedback: Customers can provide feedback on their support experience, helping the company identify areas for improvement and enhance service quality.
- Rating System: A rating system can be implemented for users to evaluate their interactions with customer support representatives.

2.9. Essential Use Cases

Use cases describe interactions between users and the system, illustrating how the application fulfills functional requirements. Below are some essential use cases for the web-based e-commerce application.



CHAPTER THREE SYSTEM ANALYSIS AND REQUIREMENT

3. INTRODUCTION

3.1. Purpose

The purpose of this document is to provide a detailed requirement analysis modeling for an e-commerce system, focusing on use cases, use case models, use case descriptions, sequence diagrams, and activity diagrams. This will facilitate a clear understanding of system functionalities and interactions.

3.2. Scope

This document covers the essential aspects of requirement analysis modeling for an e-commerce platform, including user interactions, system responses, and the overall workflow of key processes.

3.3. Definitions

- **Use Case**: A description of how a user interacts with the system to achieve a specific goal.
- **Use Case Diagram**: A visual representation of the interactions between users and the system.
- **Sequence Diagram**: A diagram that shows how objects interact in a particular scenario of a use case.
- **Activity Diagram**: A diagram that illustrates the dynamic aspects of the system and the workflow of activities.

3.4. Use Case Modeling

3.4.1. Overview of Use Cases

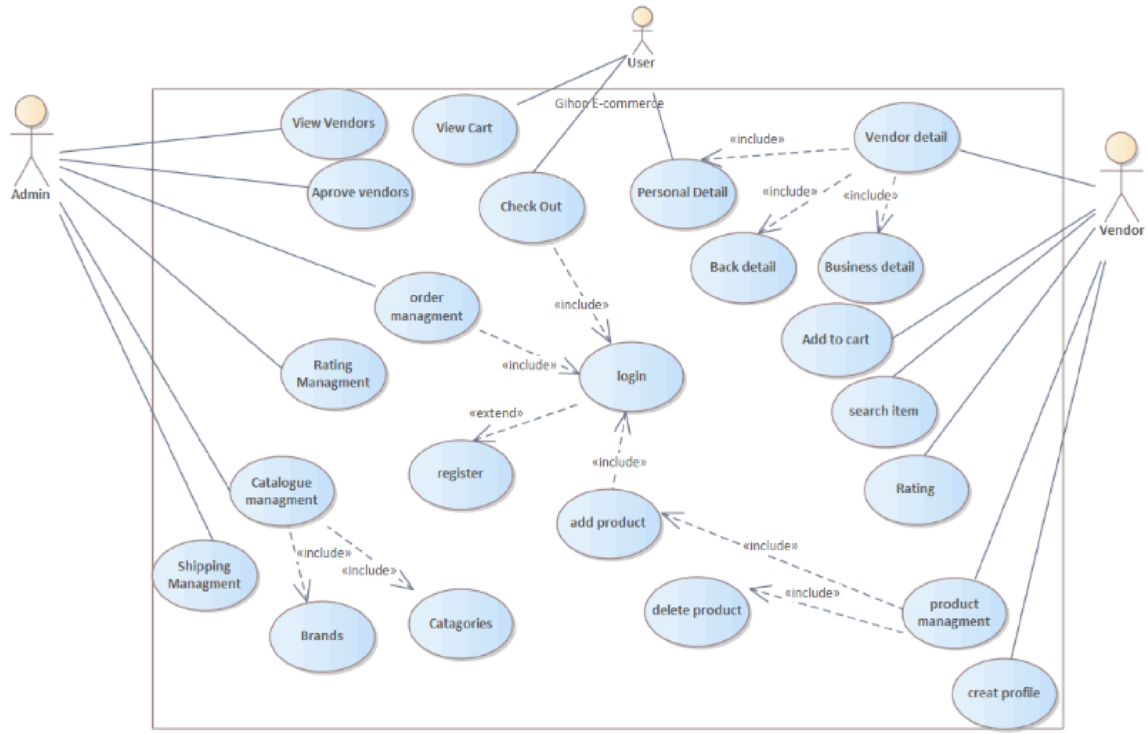
Use cases describe the functional requirements of the e-commerce system by capturing the interactions between users (actors) and the system. Each use case outlines a specific scenario in which the user achieves a goal.

- **System use case model**

In our use-case model there are a number of model elements. The most important model elements are: use cases, actors and the relationships between them.it is used to graphically depict a subset of the model to simplify communications. It is used as the primary specification of the functional requirements for our system, as the basis for analysis and design, as an input to iteration planning, as the basis of defining test cases and as the basis for user documentation.

3.4.2. Use Case Diagram

The use case diagram visualizes the relationships between actors and the system's functionalities. Below is a simplified representation of the e-commerce system use case diagram.



3.5. Use Case Descriptions

Use case description table

Table 7: use case description

Use case ID	Uc-01
Use case name	E-commerce system use cases
Actors	Admin,Client and merchant
Description	This use case outlines the processes involved in user authentication, product management, and transactions within the E-commerce system.

Use Cases Overview

Table 8: use case overview

Use case ID	Use case Name	Actor name	Description	Extend / includes
uc-01	Login	admin	Allows users (clients, vendors) to log into their accounts.	Register
uc-02	Register	users	Enables users to create a new account.	
uc-03	Reset Password		Allows users to reset their passwords if forgotten.	
uc-04	Create Profile	Vendor	Allows vendors to create their profile.	Personal Detail, Business Detail
uc-05	Add Product	vendor	Enables vendors to add new products to the system.	Product Management
uc-06	Delete Product	Vendor	Allows vendors to remove products from the system.	Product Management
uc-07	View Vendors	Admin	Allows the admin to view the list of vendors.	
uc-08	Approve Vendor	Admin	Enables admin to approve vendor registrations.	
uc-09	Rating Management	Admin	Manages ratings given by clients on products.	
uc-10	Search Item	User	Allows clients to search for items in the catalog.	

uc-11	Add to Cart	User	Enables clients to add items to their shopping cart.	
uc-12	View Cart	User	Allows clients to view items in their cart.	
uc-13	View Product Details	Admin, User	Allows users to view detailed information about specific products.	
uc-14	Checkout	User	Facilitates the checkout process for clients.	Order Management
uc-15	Order Management	Admin	Admin can manage accounts, including adding and removing. Manages customer orders.	
uc-16	Make purchase	User	Enables clients to purchase selected products.	
uc-17	Shipping Management	Admin	Handles the shipping process for orders.	

Summary of Use Cases

Use case ID	Main flow	Post conditions
uc-01	1. User enters credentials.2. The system validates and logs the user in.	The user is logged in successfully.
uc-02	1.Users fill the registration form. 2. The system validates and creates a new account.	An account is created and the user can login.

uc-03	1. User requests password reset. 2. The system sends a reset link. 3. User resets password.	Password is reset successfully.
uc-04	1. Users search for products. 2. The system displays search results. 3. User views product.	User views product details.
uc-05	1. Client fills out a request form. 2. The system sends requests to the admin.	Requests are sent and logged.
uc-06	1. The user confirms items in the cart. 2. Enter payment information. 3. Confirms order.	Order is processed and payment is received.
uc-07	1. Client submits review. 2. System records the review for the product.	Review is submitted successfully.
uc-08	1. reviews client registrations. 2. approves or rejects clients.	Client registration status is updated.
uc-09	1. selects report type. 2. The system generates and displays reports.	The report is generated successfully.
uc-10	1. Admin reviews client requests. 2. responds via the system.	Client receives a response.
uc-11	1. Admin checks payment status for orders. 2. The system displays payment details.	Payment status is verified.

uc-12	1. enters product details. 2. The system adds products to inventory.	Product is added successfully.
uc-13	1. The user selects a product. 2. The system displays detailed product information.	User views detailed product information.
uc-14	1. Admin accesses system settings. 2. Admin adjusts configurations as needed.	System settings are updated.
uc-15	1. Admin views list. 2. Admin adds/removes s as needed.	management is updated.
uc-16	1. User adds product to cart.2. Proceeds to checkout. 3. Confirms purchase and payment.	Purchase is completed and confirmation sent.
uc-17	1. Admin accesses reports section. 2. The system displays reports on sales and activity.	Reports are viewed.
uc-18	1. The user clicks logout. 2. The system logs the user out and ends the session.	The user logged out successfully.

3.6. Sequence Diagrams

3.6.1. Overview of Sequence Diagrams

Sequence diagrams illustrate how objects in the system interact with each other in a specific use case scenario, detailing the order of messages exchanged.

We used Sequence diagrams to represent or model the flow of messages, events and actions between the objects or components of our system. We represent the time in the vertical direction showing the sequence of interactions of the header elements, which are displayed horizontally at the top of the diagram. We used Sequence Diagrams primarily to design, document and validate the architecture, interfaces and logic of the system by describing the sequence of actions that need to be performed to complete a task or scenario. Our sequence diagrams are useful design tools because it provides a dynamic view of our system behavior which can be difficult to extract from static diagrams or specifications

3.6.2. Sequence Diagram for User Registration

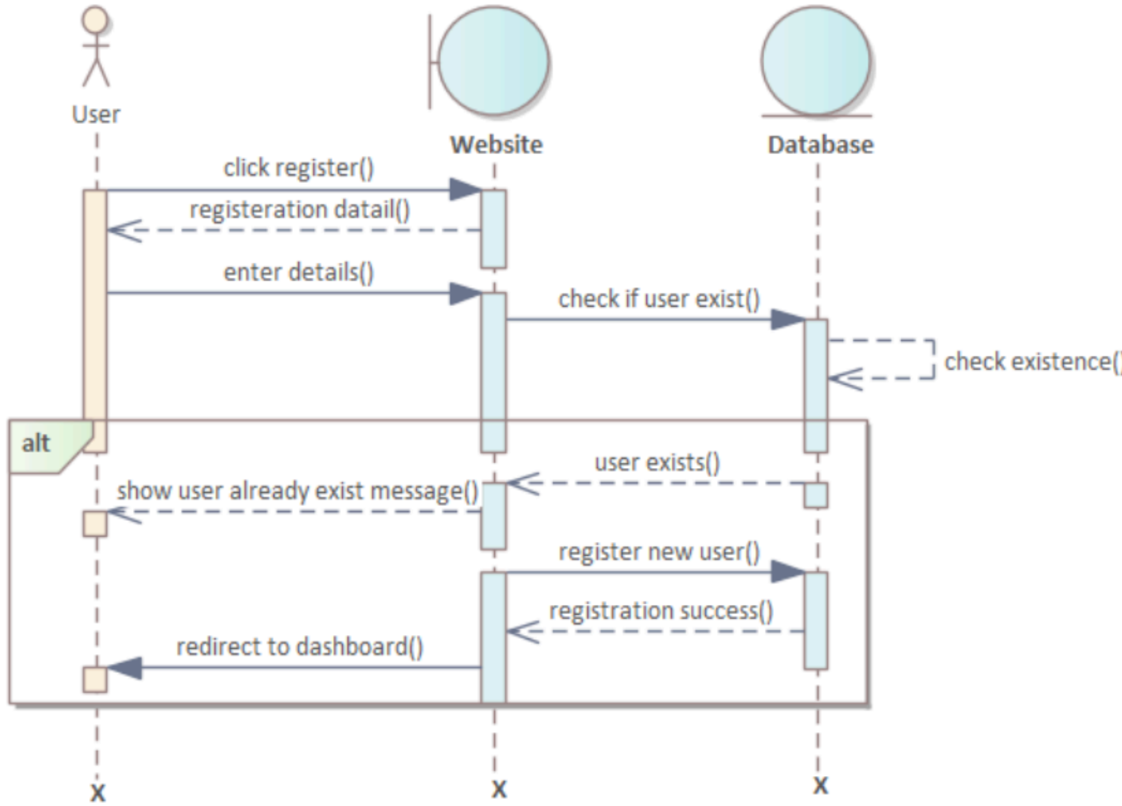


Figure 3: sequence diagram for registration

3.6.4 Sequence diagram for checkout process

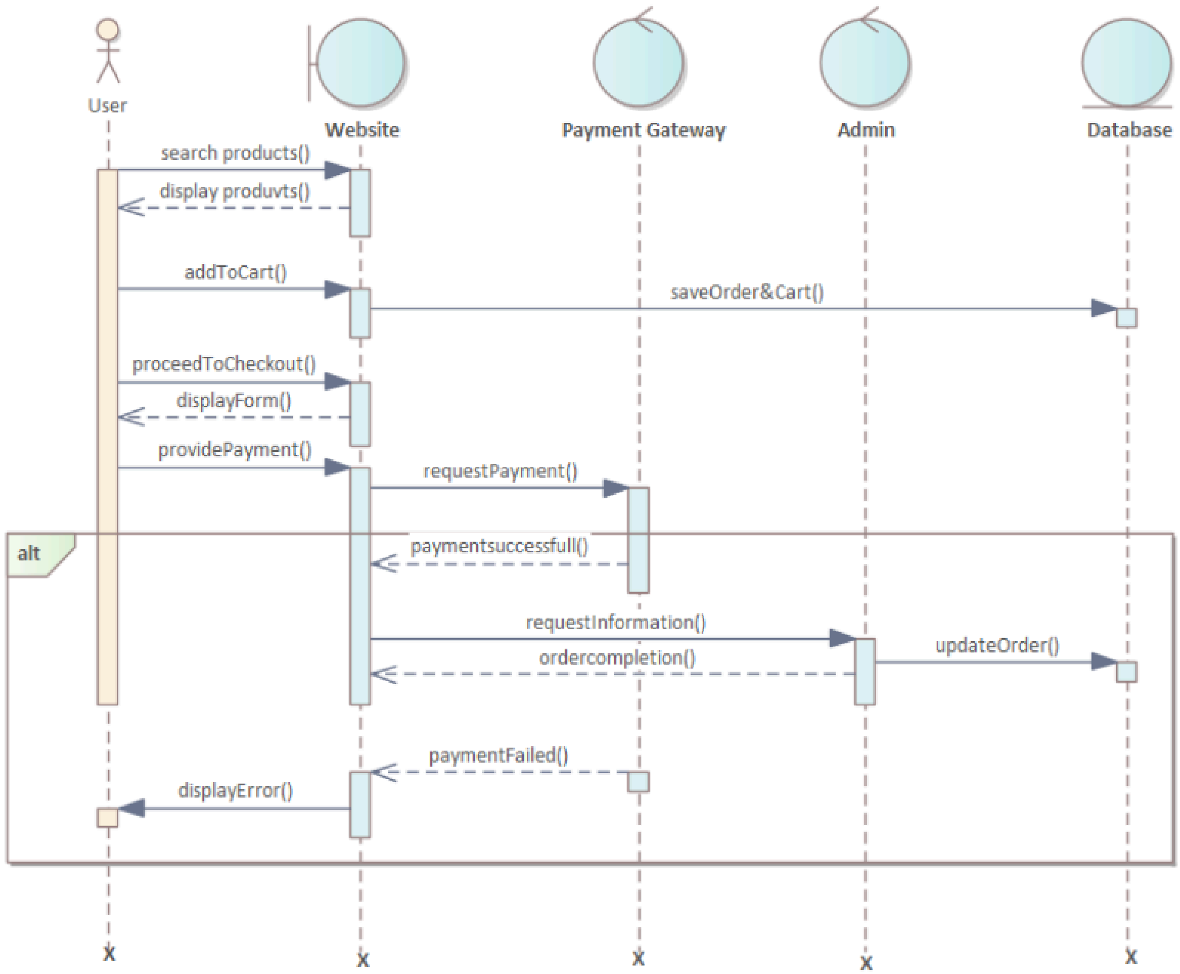
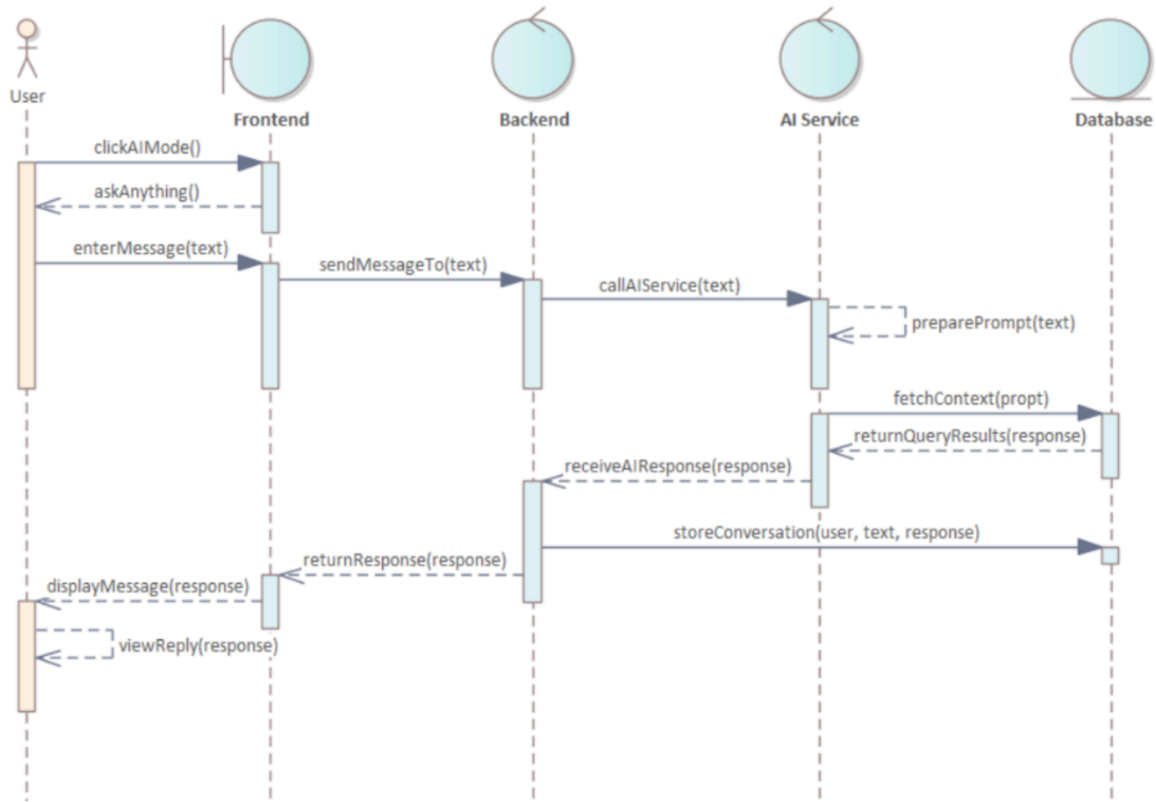


Figure 4: sequence diagram for checkout process

3.6.5 sequence diagram for chatAI process



3.6.6 sequence diagram for chatAI process

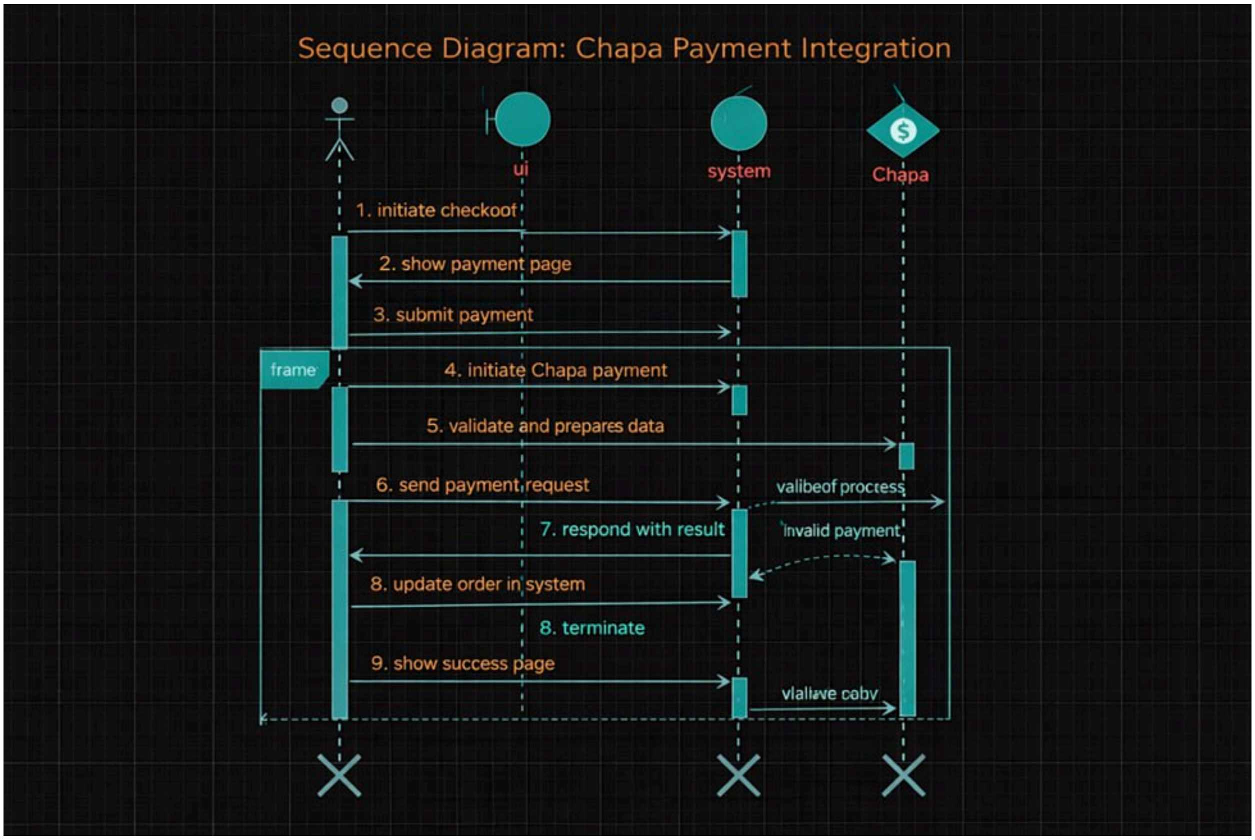


Fig 7 Chapa payment integrations

3.7 Activity Diagrams

3.7.1 Overview of Activity Diagrams

Activity diagrams depict the workflow of activities within a process, showing the sequence of actions and decisions.

3.7.2 Activity Diagram model for user registration

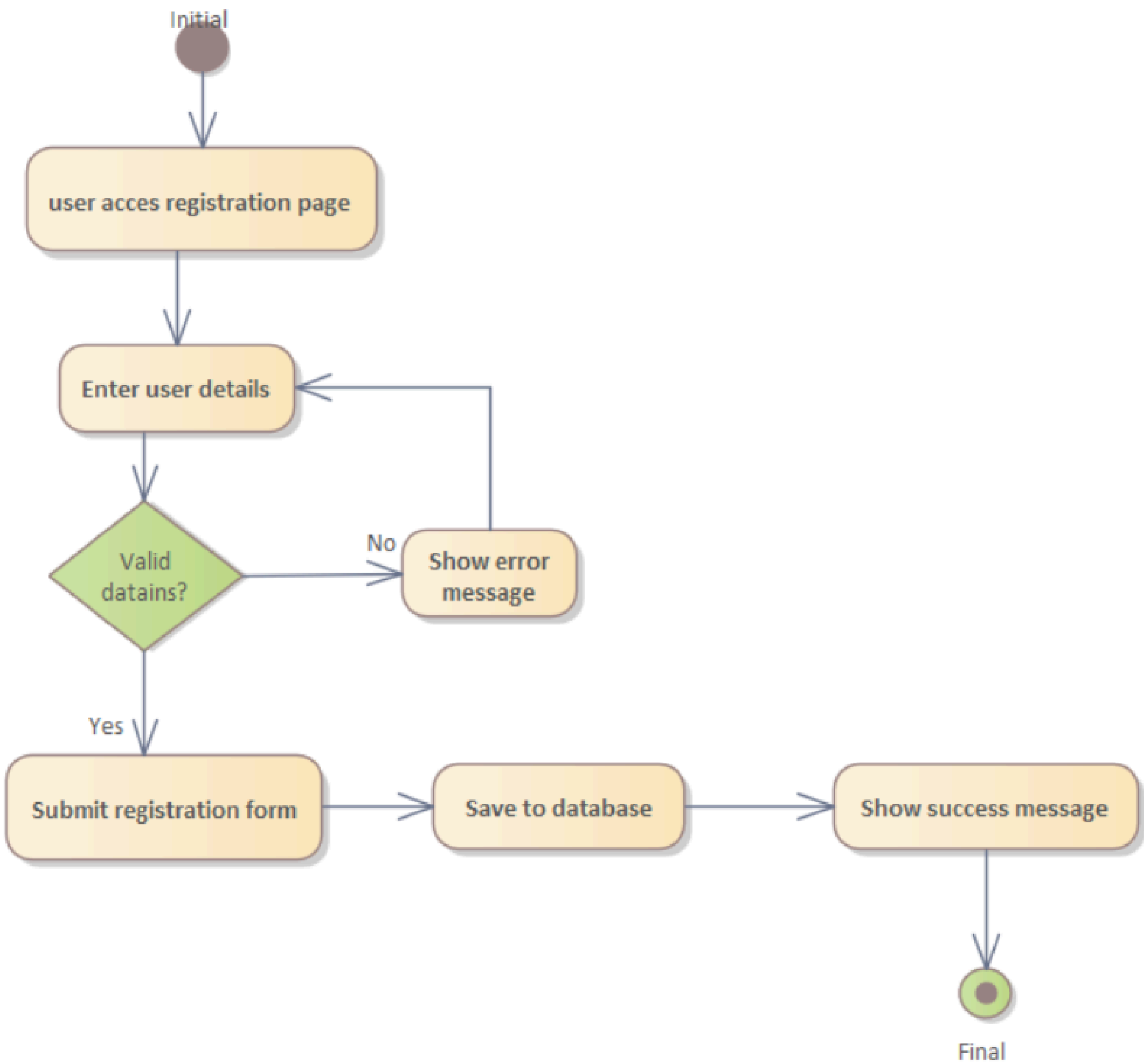


Figure 5:activity diagram for registration

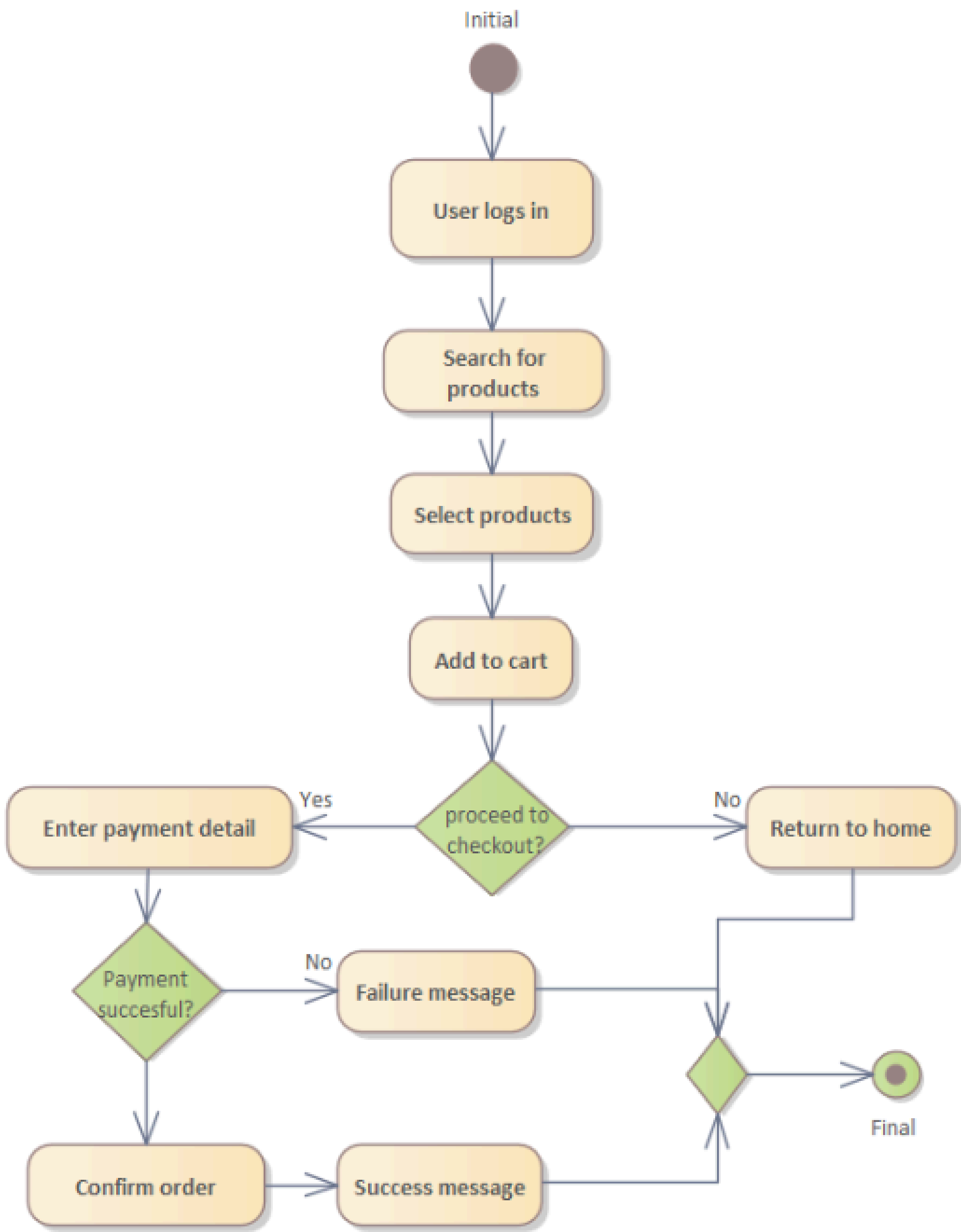

Initial
User logs in
Search for products
Select products
Add to cart
proceed to checkout?
Yes
No
Enter payment detail
Return to home
Payment succesful?
No
Failure message
Final
Confirm order
Success message

Figure 6: activity diagram for checkout

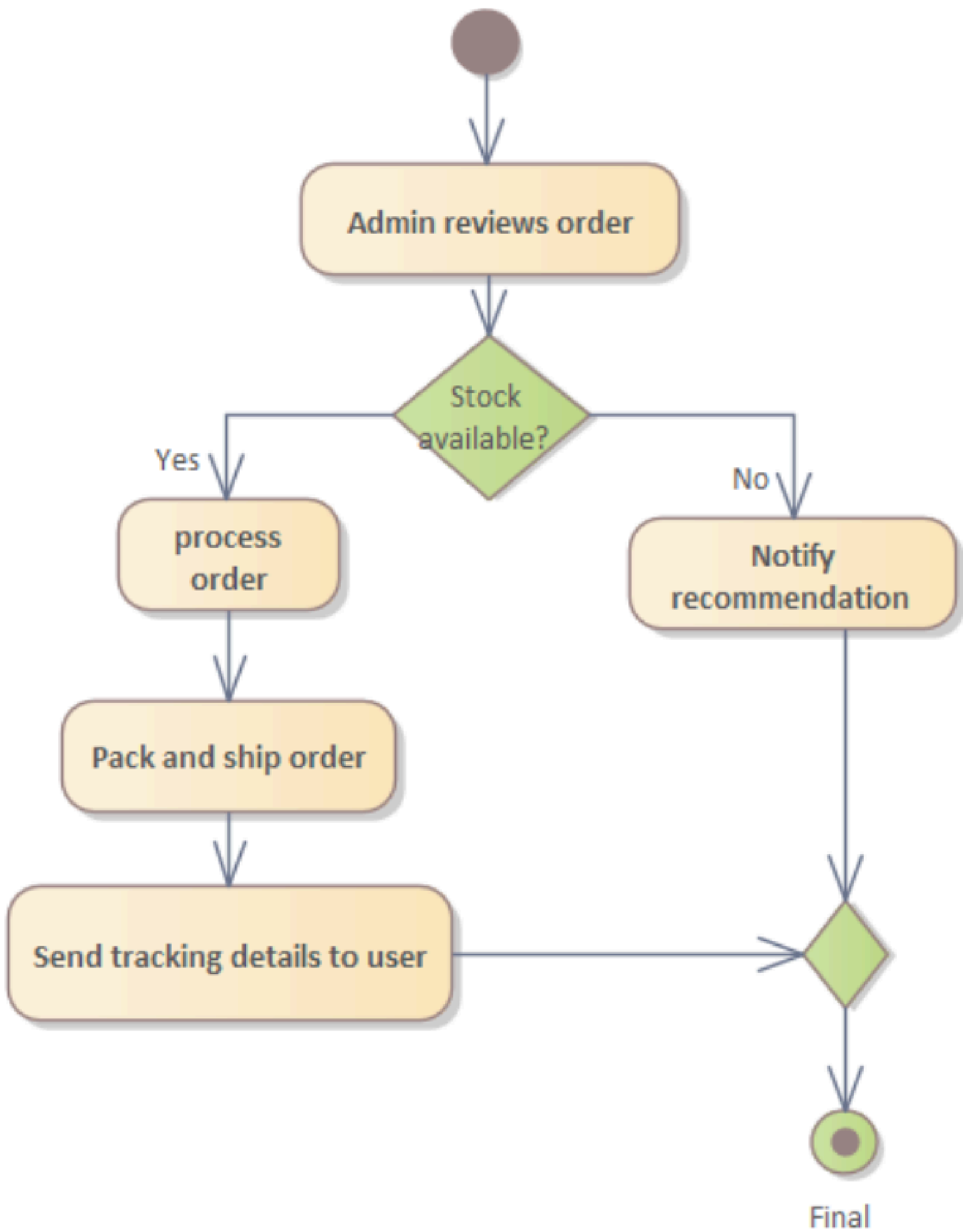


Figure 6: activity diagram for order approving

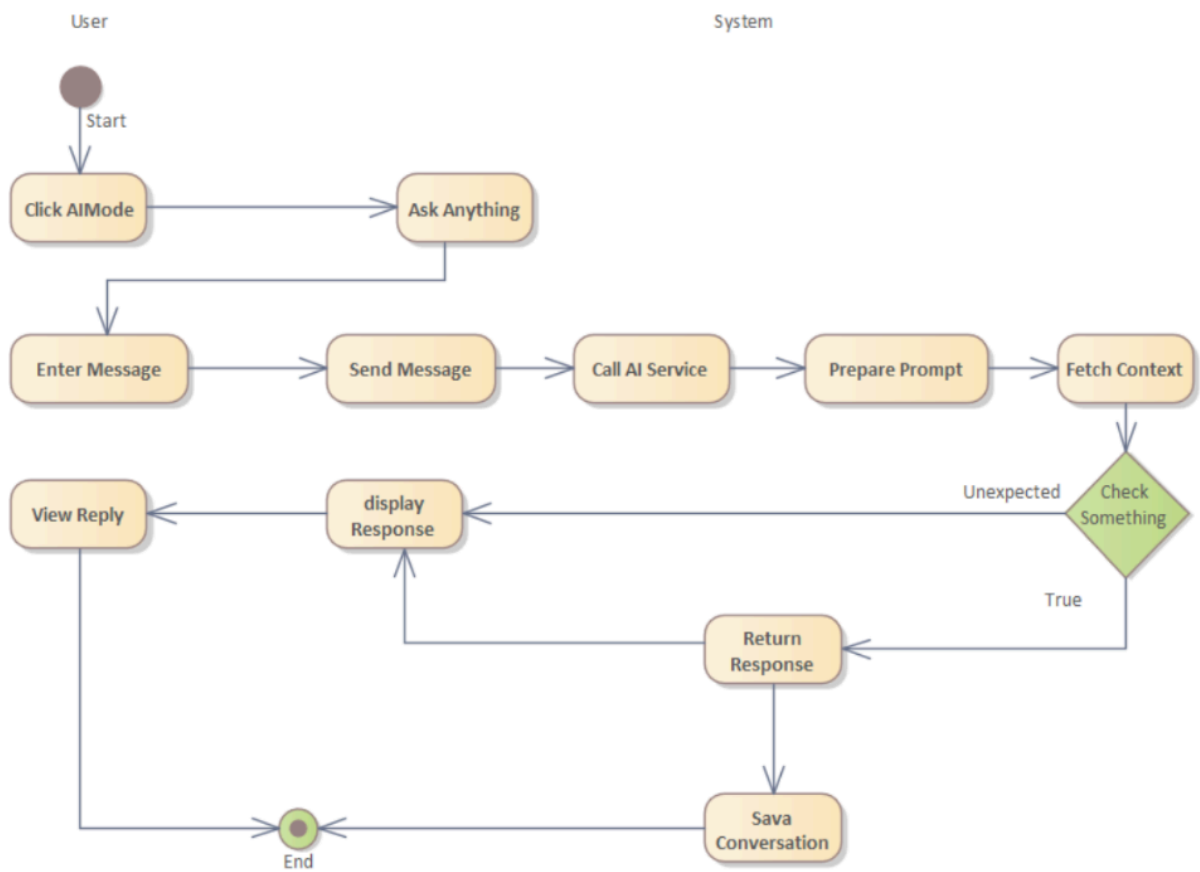
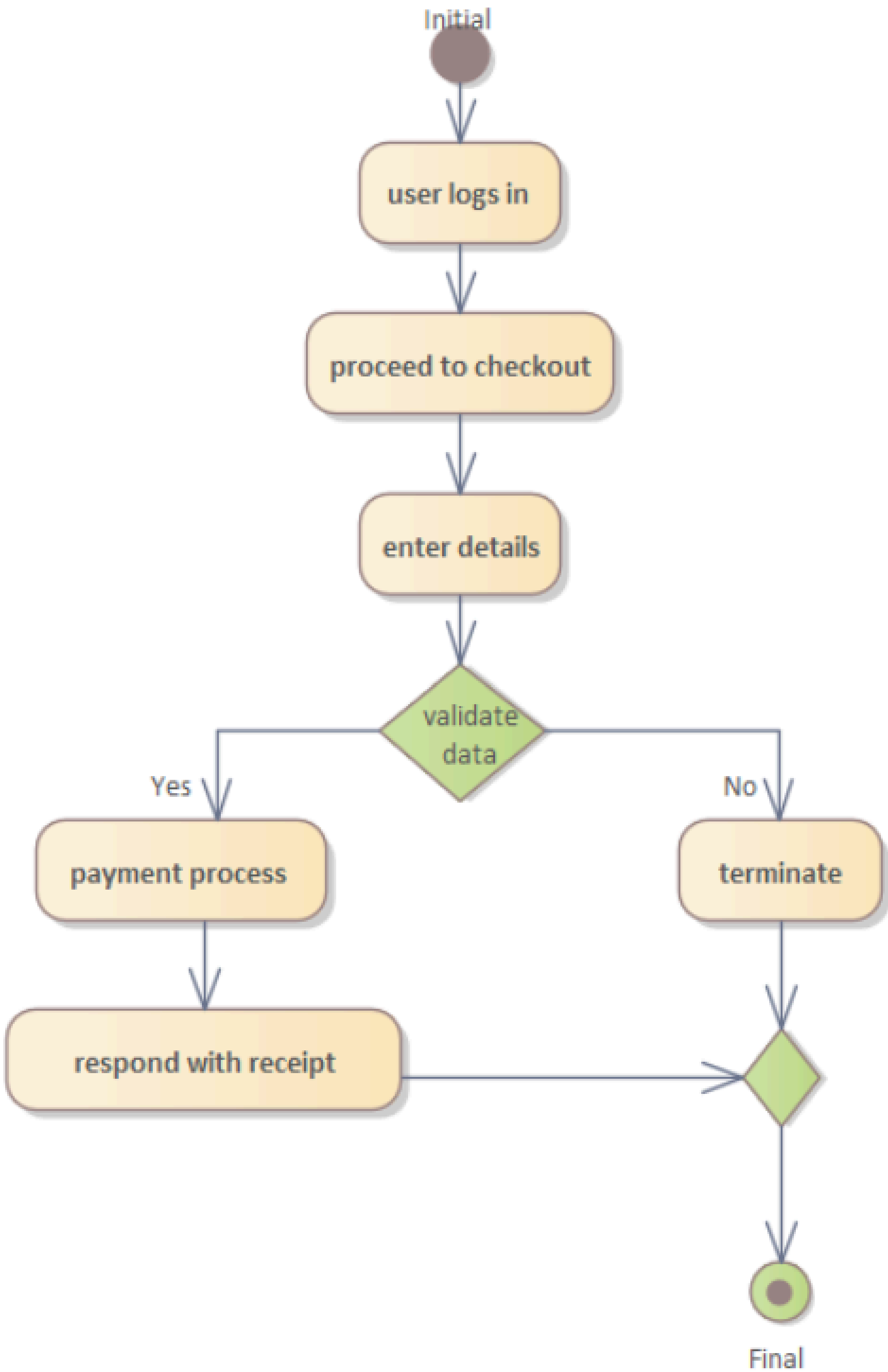


Figure7: Chatting with AI activity diagram

Figure 8: Payment with chapa activity diagram



CHAPTER FOUR SYSTEM DESIGN

4. INTRODUCTION

This document serves as a comprehensive guide to the system design of a web-based e-commerce application. It outlines the architecture, components, and interactions that will facilitate the effective functioning of the platform. By focusing on various design elements such as use case modeling, class diagrams, and deployment strategies, this document aims to provide clear insights into how the e-commerce system will operate and meet the needs of both users and administrators. The primary purpose of this system design document is to ensure that all stakeholders developers, project managers, business analysts, and clients have a unified understanding of the system's architecture and functionalities. This shared understanding is crucial for aligning development efforts with business goals, ensuring that the resulting application not only meets functional requirements but also adheres to non-functional requirements such as security, performance, and scalability.

4.1. Significance of System Design in E-Commerce Applications

System design plays a pivotal role in the development of e-commerce applications due to the unique challenges and requirements they present. E-commerce platforms must effectively manage a variety of functionalities, including user accounts, product catalogs, payment processing, and order management, all while providing a seamless user experience. Here are several key reasons why system design is particularly significant in this context

4.2. Proposed system architecture

The architecture for the proposed e-Commerce system follows a multi-tier approach, ensuring scalability, security, and seamless integration of various components. The architecture is divided into three primary layers:

4.2.1. Presentation Layer (Frontend)

The user-facing interface where clients, merchants, and s interact with the system. It will be built using HTML and Js for dynamic and responsive user experiences, integrated with internal CSS for modern, scalable styling.

Components:

User registration and login pages.

Product search, listing, and purchase features.

Admin and merchant dashboards for product management.

Client-side validations for enhanced usability.

4.2.2 Application Layer (Backend)

The core of the system that handles business logic, data processing, and API management. This layer will use PHP to manage server-side logic, ensuring efficient communication between the frontend and the database.

Components:

RESTful APIs for product, user, order, and management.

4.2.3. Data Layer (Database)

The storage and retrieval system for all application data. The database will use MySQL for structured data storage, ensuring relational integrity for all user, product, order, and transaction records.

Components:

User database for clients, merchants, and admin

Product catalog and inventory management tables.

Order history, transaction, and payment data.

4.3. System Design Model

A Low-Level Design (LLD) model for an e-commerce platform focuses on the detailed architecture of the system components, including classes, methods, and interactions. Below is a conceptual overview of various components involved in the LLD for an e-commerce system.

4.3.1. Class Diagram

A class diagram provides a high-level overview of the structure of a system by modeling its key entities, their attributes, behaviors, and relationships. It illustrates how different parts of the system interact with one another, defining the roles and responsibilities of each component. The diagram establishes a blueprint for how data flows between different entities, how actions are performed, and how these components are related or dependent on each other. It serves as an essential tool for developers, allowing them to visualize the system's architecture and ensure that the design is both scalable and maintainable.

The Class diagram captures the logical structure of the system; the classes and things that make up the model. It is a static model, describing what exists and what and behavior it has, rather than how something is done. Class diagrams are attributes most useful to illustrate relation- ships between classes and interfaces.

4.3.2. Class Diagram model for the entire system

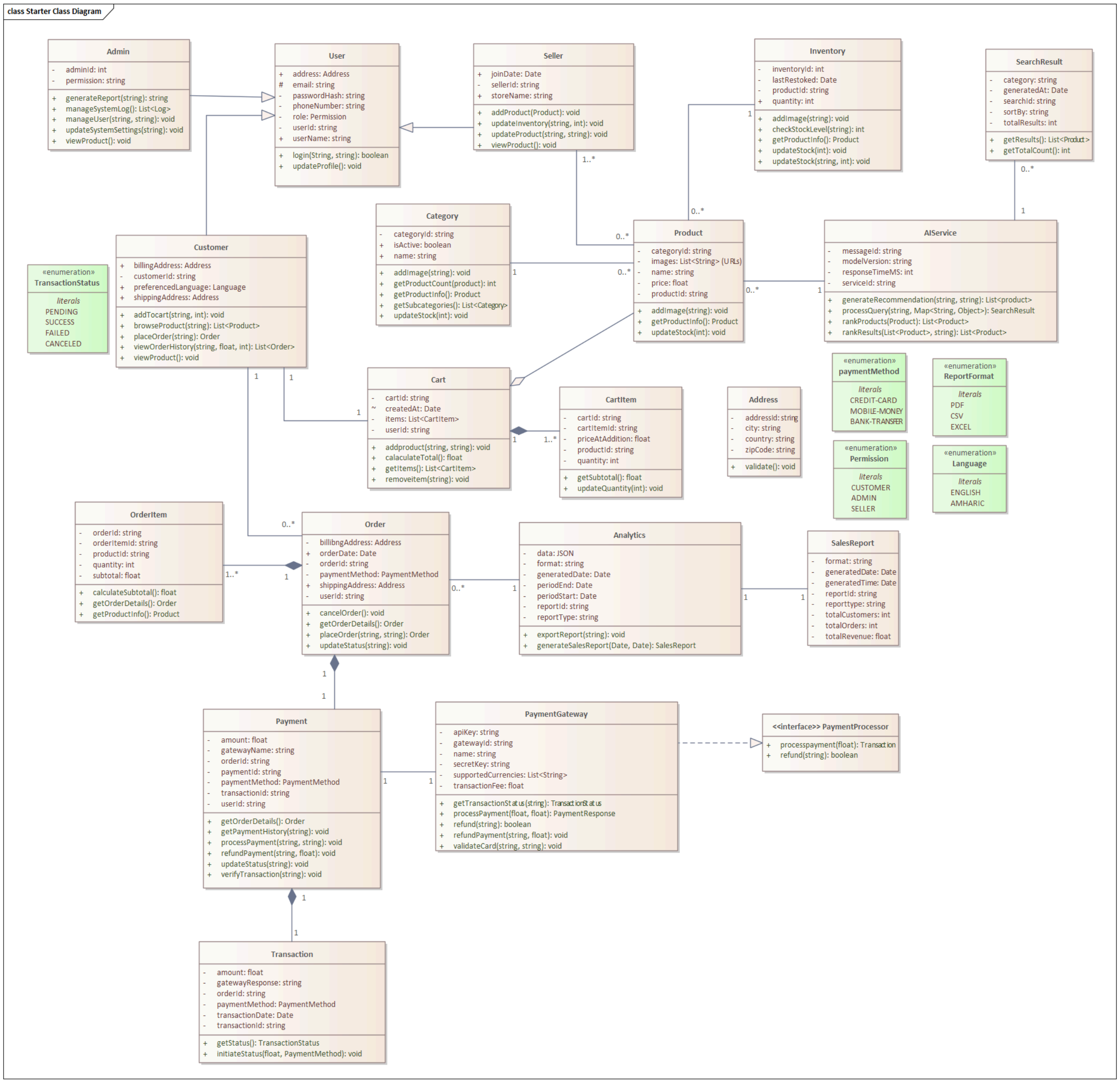
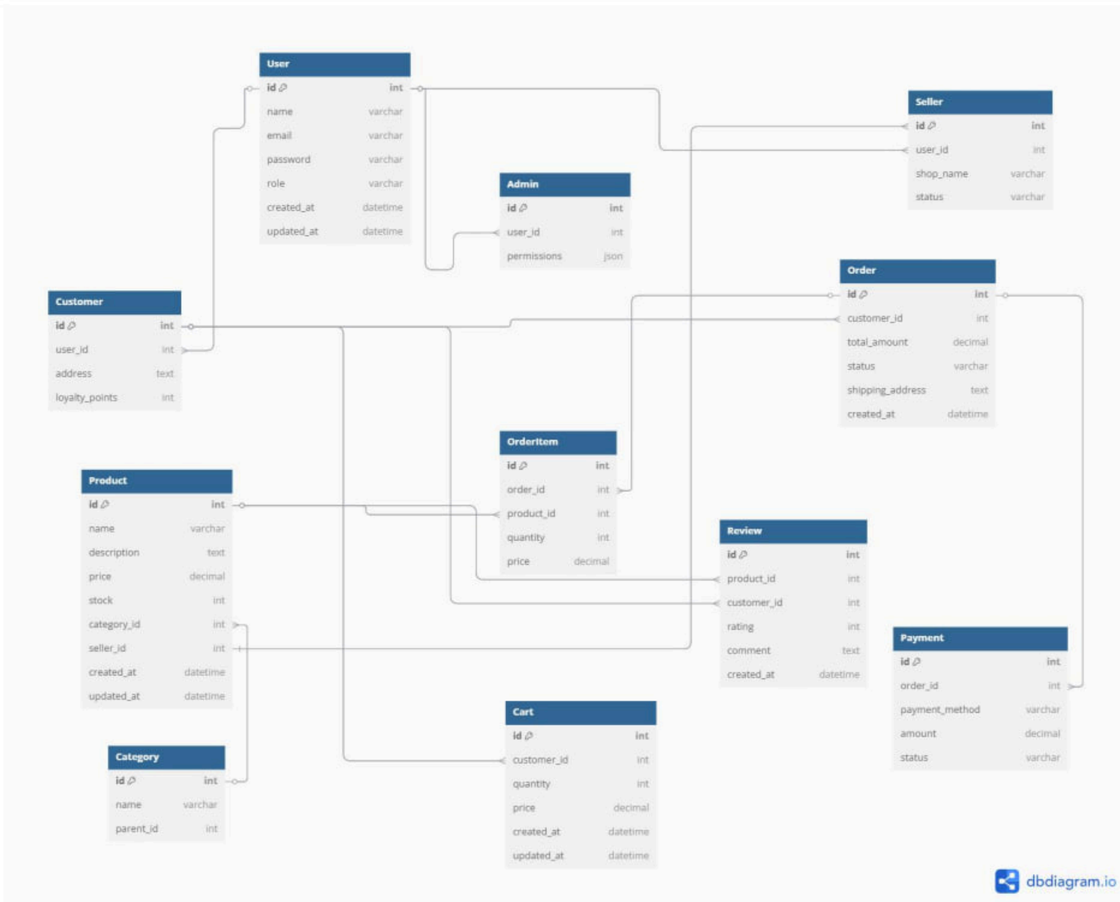


Figure 7: class diagram model

4.4. Database Design

database design for an e-commerce platform that includes the entities you've mentioned:- Users, Products, Categories, Orders, OrderItems, ShoppingCart, CartItems, and Payments. Each entity is represented as a table with its attributes and relationships.



CHAPTER FIVE: IMPLEMENTATION AND TESTING

5.1. Overview

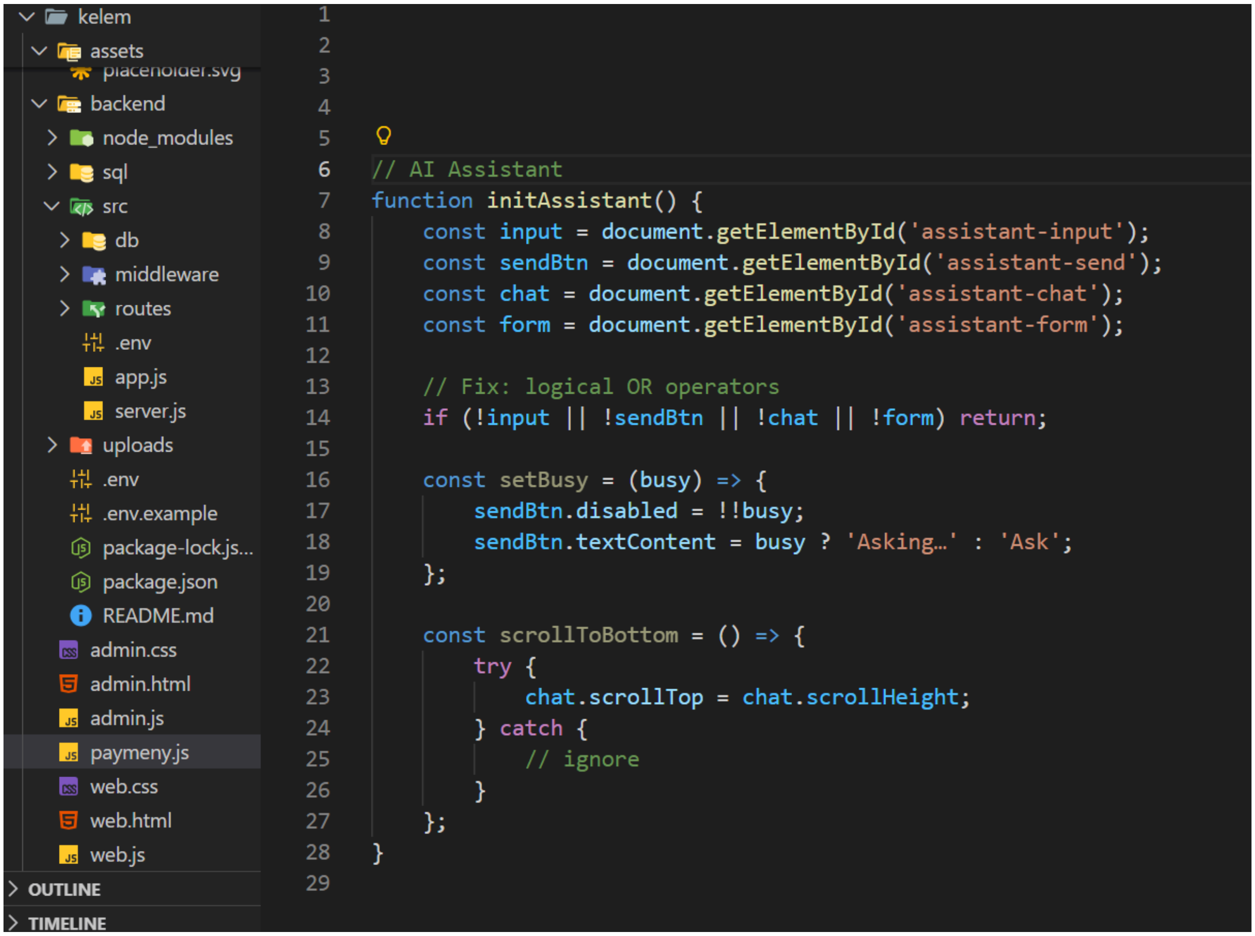
In this portion of the chapter, we used to describe the implementation and testing phase of the system.

5.2. Implementation

Implementation refers to the Coding of all documents gathered starting from requirement analysis to Design phase. In the implementation phase all the programs are written, a database is created, user operational documents are written, users are trained, and the system tested with operational data.

As mentioned earlier we used javascript to develop our system. In the following we will show the sample codes of our system functionalities to apply for login.

Sample code for AI




```javascript
// AI Assistant
function initAssistant() {
    const input = document.getElementById('assistant-input');
    const sendBtn = document.getElementById('assistant-send');
    const chat = document.getElementById('assistant-chat');
    const form = document.getElementById('assistant-form');

    // Fix: logical OR operators
    if (!input || !sendBtn || !chat || !form) return;

    const setBusy = (busy) => {
        sendBtn.disabled = !!busy;
        sendBtn.textContent = busy ? 'Asking…' : 'Ask';
    };

    const scrollToBottom = () => {
        try {
            chat.scrollTop = chat.scrollHeight;
        } catch {
            // ignore
        }
    };
}
```

web.js paymeny.js

kelem > paymeny.js > ...

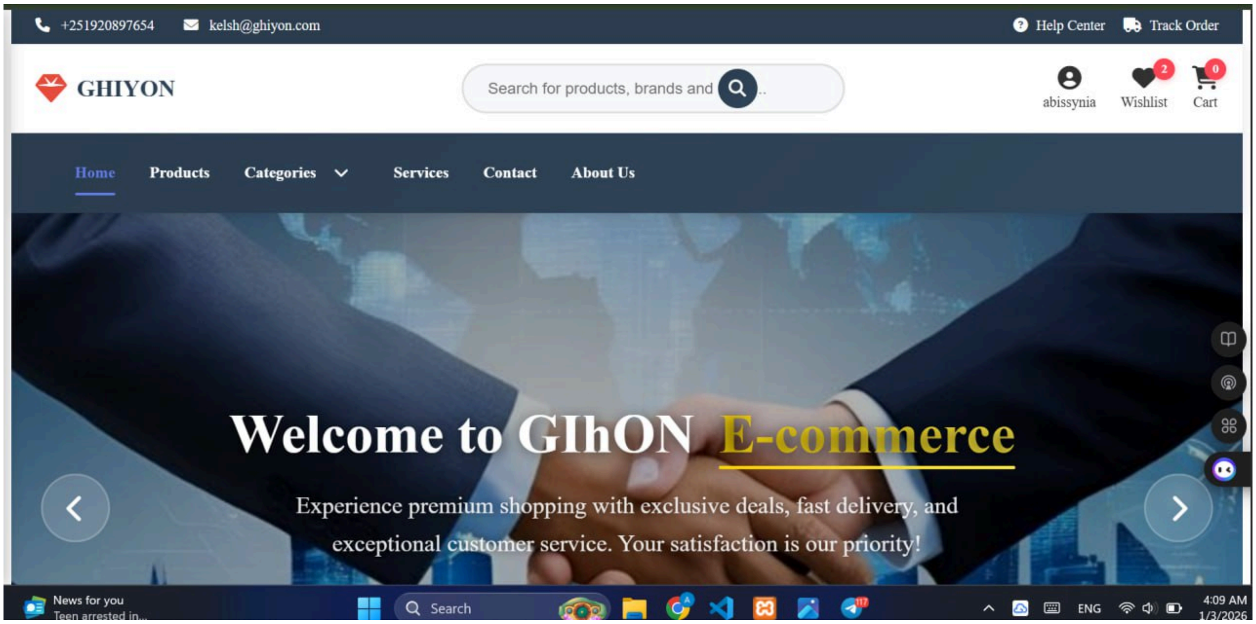
```
// Get selected payment method
const payment = document.querySelector('input[name="payment-method"]:checked');
const paymentMethod = payment ? String(payment.value) : 'cod';

if (paymentMethod === 'chapa' && !token) {
    showNotification('Please login to pay with Chapa.', 'error');
    showLogin();
    return;
}

try {
    // Always create order first (server computes totals)
    const order = await apiJson('/api/orders', 'POST', {
        items,
        shippingFee: items.length > 0 ? SHIPPING_FEE : 0,
        paymentMethod
    });

    // ---------------- CHAPA PAYMENT ----------------
    if (paymentMethod === 'chapa') {
        const firstName = String(
            document.getElementById('checkout-first-name')?.value || ''
        ).trim();

        const lastName = String(
            document.getElementById('checkout-last-name')?.value || ''
        ).trim();

        const email = String(
            document.getElementById('checkout-email')?.value || ''
```

```javascript
).trim();

const txRef = `order_${order.id}_${Date.now()}`;

// Save pending payment info
try {
    localStorage.setItem(
        PENDING_CHAPA_KEY,
        JSON.stringify({
            txRef,
            orderId: Number(order.id),
            at: Date.now()
        })
    );
} catch {
    // ignore storage errors
}

const init = await apiJson('/api/payments/chapa/initialize', 'POST', {
    amount: Number(order.total),
    currency: 'ETB',
    email,
    firstName,
    lastName,
    txRef,
    orderId: Number(order.id)
});

// Clear server cart (ignore failures)
try { await apiJson('/api/cart', 'DELETE'); } catch {}
```

```
    // Clear server cart (ignore failures)
    try { await apiJson('/api/cart', 'DELETE'); } catch {}

    // Clear local cart before redirect
    cart = [];
    serverCartItems = null;
    updateCartCount();
    updateCartDisplay();
    saveCart();
    updateCheckoutTotals();
    closeCheckout();

    showNotification('Redirecting to Chapa payment...', 'success');
    window.location.href = String(init.checkoutUrl);
    return;
}

// ---------------- CASH ON DELIVERY ----------------
try { await apiJson('/api/cart', 'DELETE'); } catch {}

cart = [];
serverCartItems = null;
updateCartCount();
updateCartDisplay();
saveCart();
updateCheckoutTotals();
closeCheckout();

showNotification(`Order placed successfully! Order #${order.id}`, 'success');
```

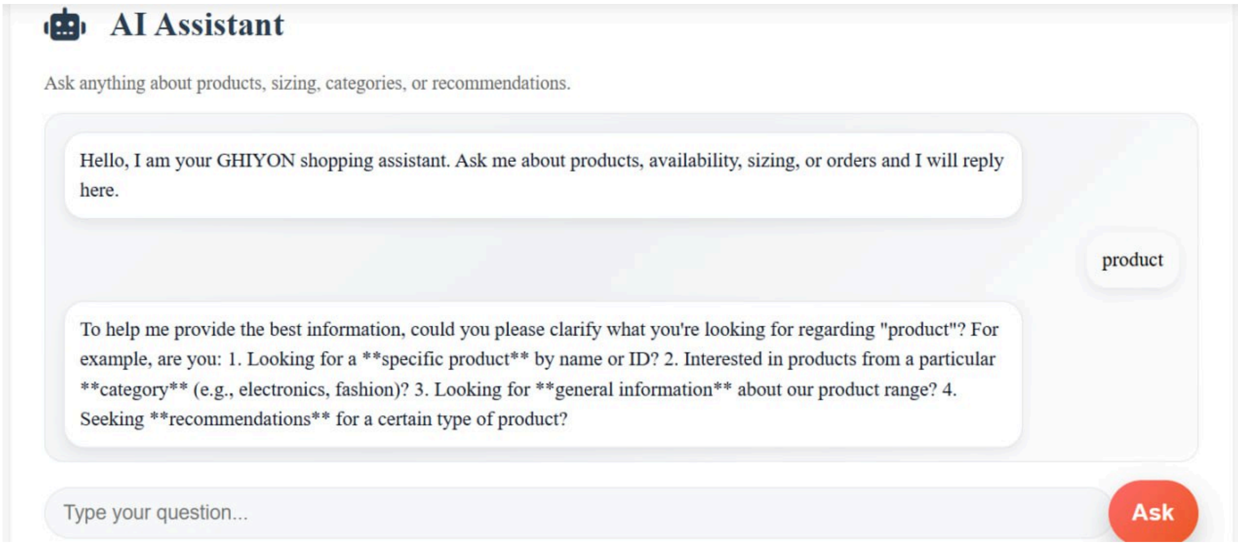
5.2.1. System Screenshots

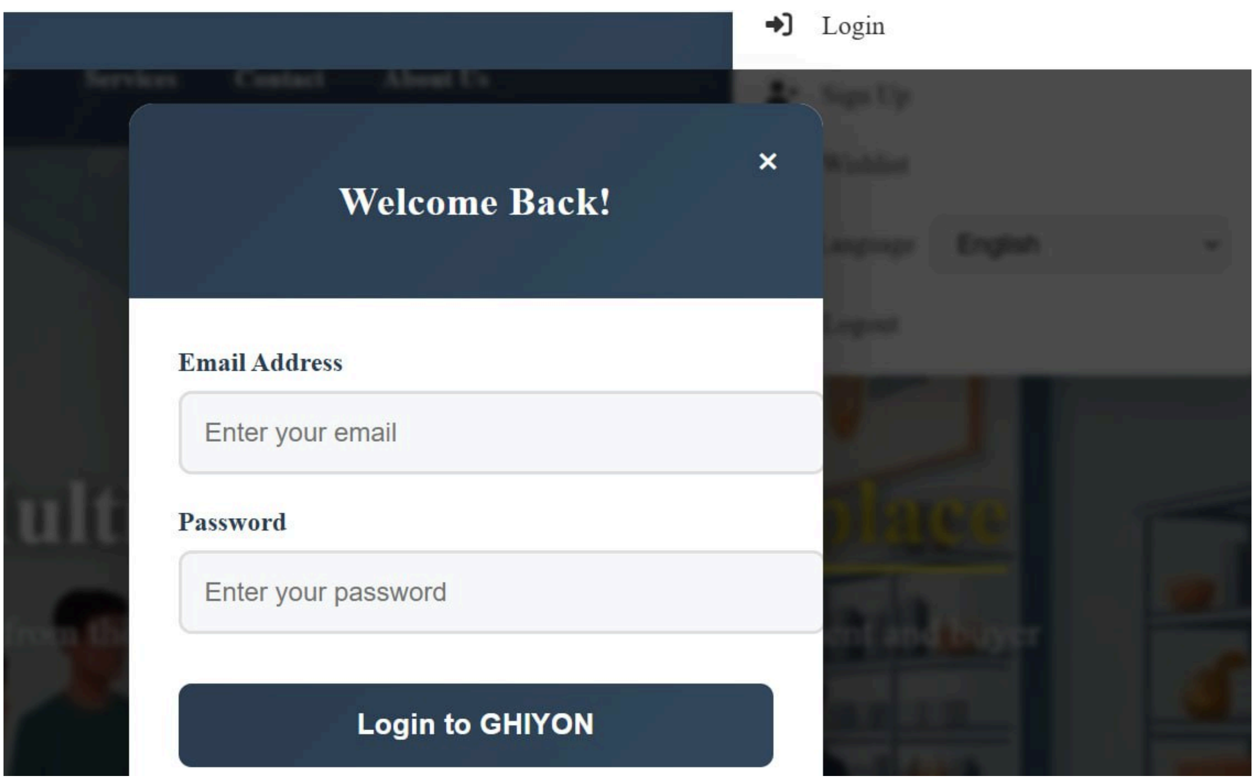
The following pictures show some screenshots of our system (GIHON E-COMMERCE).

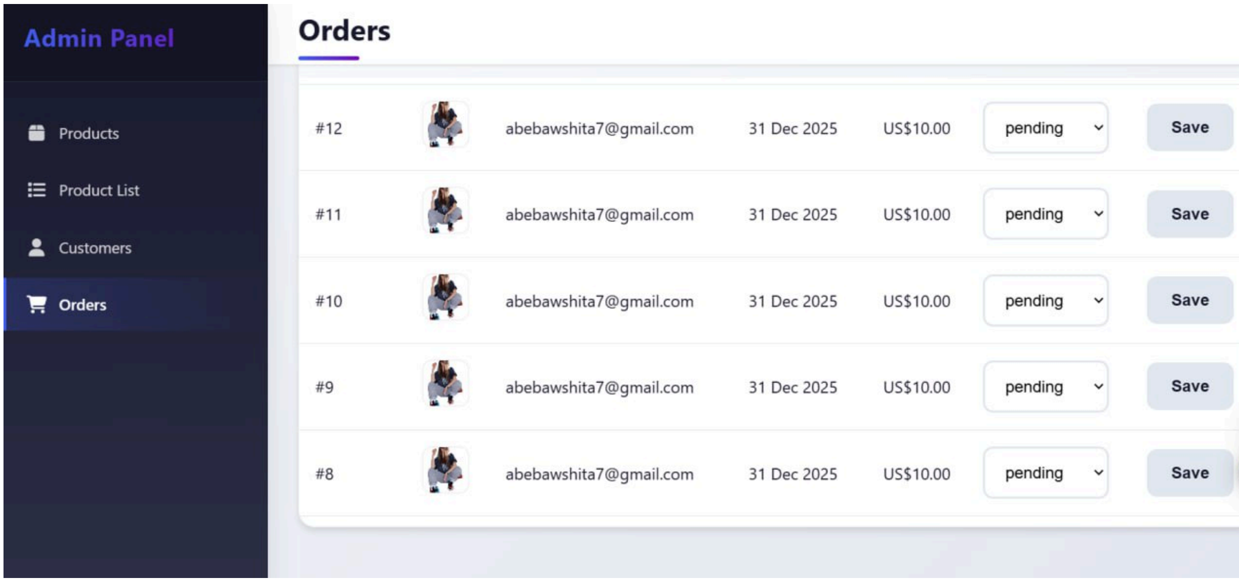


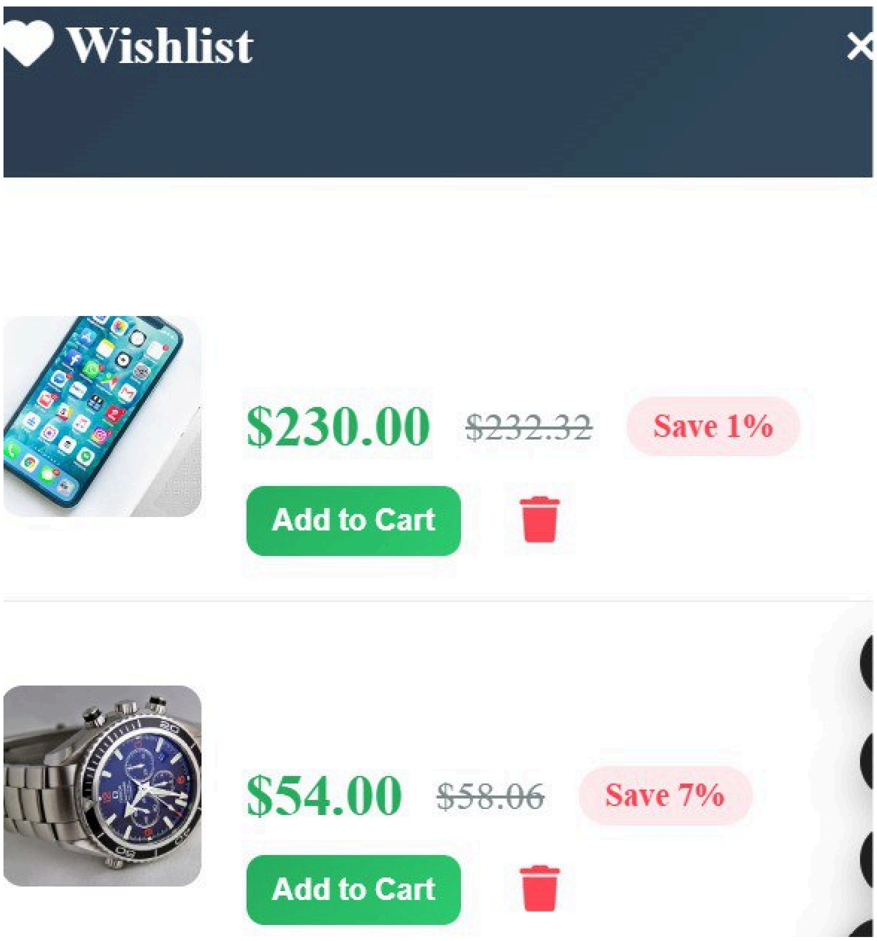
Products

AI page that is integrated with my system screenshot sample

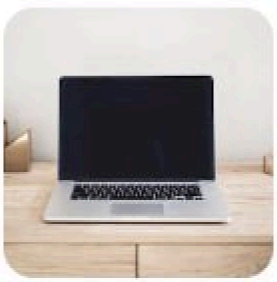

AI Assistant
Ask anything about products, sizing, categories, or recommendations.
Hello, I am your GHIYON shopping assistant. Ask me about products, availability, sizing, or orders and I will reply here.
product
To help me provide the best information, could you please clarify what you're looking for regarding "product"? For example, are you: 1. Looking for a **specific product** by name or ID? 2. Interested in products from a particular **category** (e.g., electronics, fashion)? 3. Looking for **general information** about our product range? 4. Seeking **recommendations** for a certain type of product?
Type your question...
Ask


Login
×
Welcome Back!
Email Address
Enter your email
Password
Enter your password
Login to GHIYON


Admin Panel
Products
Product List
Customers
Orders
Orders
#12 abebawshita7@gmail.com 31 Dec 2025 US$10.00 pending Save
#11 abebawshita7@gmail.com 31 Dec 2025 US$10.00 pending Save
#10 abebawshita7@gmail.com 31 Dec 2025 US$10.00 pending Save
#9 abebawshita7@gmail.com 31 Dec 2025 US$10.00 pending Save
#8 abebawshita7@gmail.com 31 Dec 2025 US$10.00 pending Save


Wishlist
$230.00
$232.32
Save 1%
Add to Cart
$54.00
$58.06
Save 7%
Add to Cart

Your Shopping Cart ×



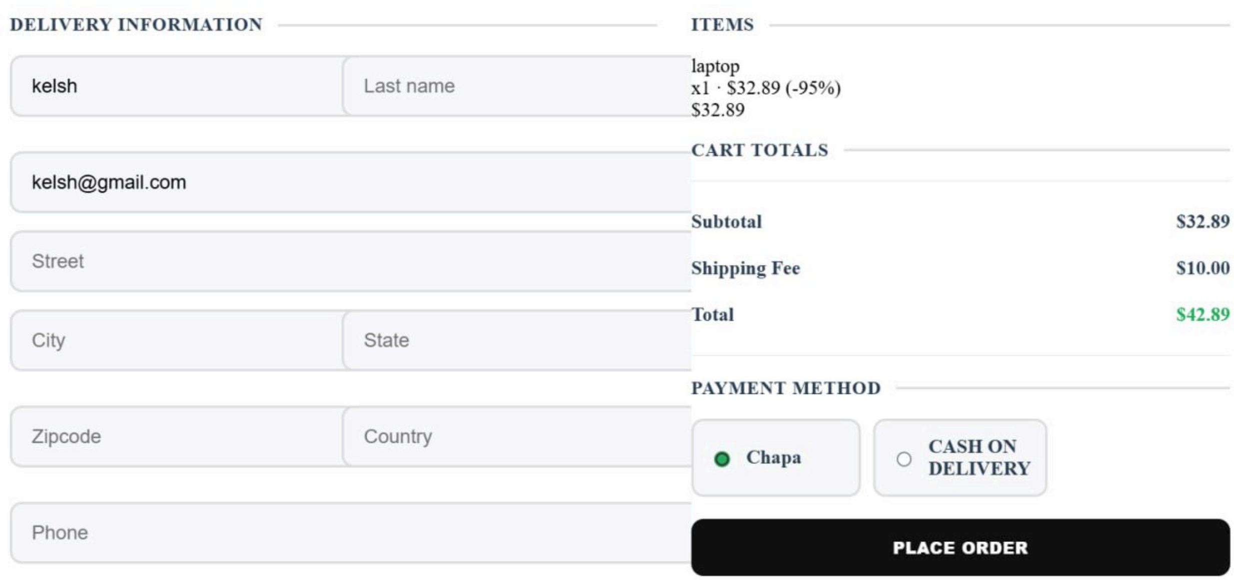
laptop

$32.89 (-95% off)

- 1 +

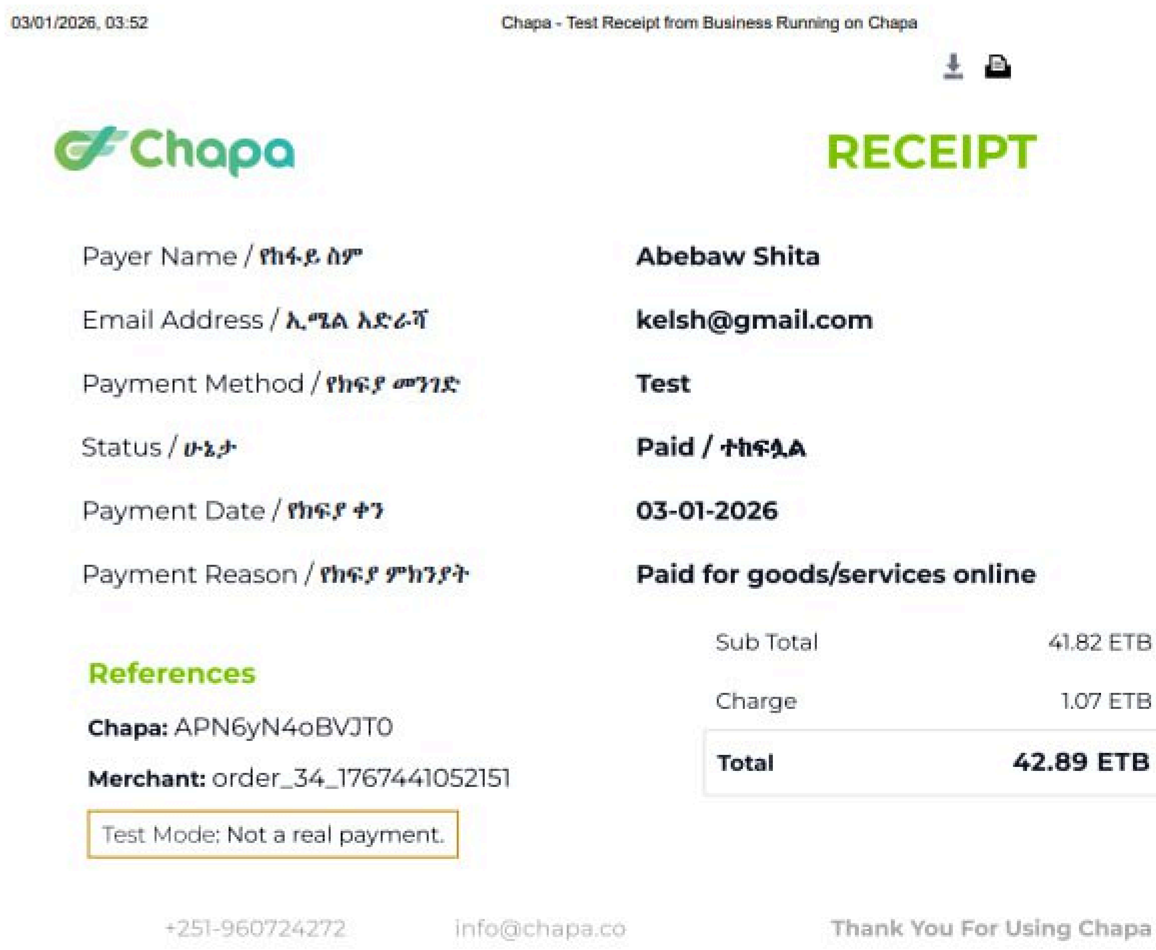
Total: **$32.8**

Proceed to Checkout



Admin Dashboard

Example samples of the chap payment reception



03/01/2026, 03:52 Chapa - Test Receipt from Business Running on Chapa

Chapa

RECEIPT

Payer Name / የከፋይ ስም	**Abebaw Shita**
Email Address / ኢሜል አድራሻ	**kelsh@gmail.com**
Payment Method / የክፍያ መንገድ	**Test**
Status / ሁኔታ	**Paid / ተከፍሏል**
Payment Date / የክፍያ ቀን	**03-01-2026**
Payment Reason / የክፍያ ምክንያት	**Paid for goods/services online**

References

Chapa: APN6yN4oBVJT0

Merchant: order_34_1767441052151

Test Mode: Not a real payment.

Sub Total	41.82 ETB
Charge	1.07 ETB
Total	**42.89 ETB**

+251-960724272 info@chapa.co Thank You For Using Chapa

Footer

5.3. Testing

Testing is the process of evaluating a system or its component(s) with the intent to find that whether it satisfies the specified requirements or not. This activity results in the actual, expected and difference between their results. Testing is executing a system in order to identify any gaps, errors or missing requirements in contrast to the actual desire or requirements.

5.3.1. Unit Testing

every module of the System is separately tested. I.e., the team tests every module by applying some selection mechanism. Through this mechanism every module gets tested. If an error occurs correction will be taken without affecting another module.

Unit testing gives stress on modules independently of one another, to find errors. This helps the tester in detecting errors in coding and logic that are contained within that module alone. The errors resulting from the interaction between modules are initially avoided.

5.3.2. Integration Testing

In this testing part, all the modules will be combined together and tested it for its fitness with each other and with the system's functionality. If error occurs in combining them, the module problems will be identified and re combined.

5.3.3. System Testing

System test ensures that the entire integrated software system meets requirements. It tests a configuration to insure known and predictable results. System testing is based on process description and flows, emphasizing pre-driven process links and integration points. In essence System testing is not about checking the individual parts of design, but about checking the system as a whole. In effect it is one giant component. System testing ensures the following have been met correctly. These are**:**

·*Functional requirements*

·*Non-Functional requirements*

5.4. User Guide and Training

5.4.1. User Guidance

A comprehensive user guide was not required, as the system is designed to be intuitive and user-friendly. However, minimal documentation is provided on the website to assist users in navigating key functionalities.

5.4.2 Training

No extensive training sessions were necessary. The system is straightforward for both administrators and end-users. However, brief training sessions were conducted for:

Administrators: Managing product listings, orders, and user accounts

Users: Navigating the platform, placing orders, and making payments

5.5. System Deployment and Installation

5.5.1. Installation Process

Since the e-commerce system is web-based, no installation is required on individual machines. Users can access the platform through a web browser via a designated URL. The only necessary setup is hosting the system on a web server and configuring the database.

5.5.2. Startup Strategy

Upon accessing the platform, users must log in with their credentials to access personalized services. The system is structured as follows:

Administrators: Require authentication to manage system functionalities.

Registered Users: Have access to product catalogs, shopping carts, and order tracking.

Guest Users: Can browse available products but must register to make purchases.

CHAPTER SIX: CONCLUSION AND RECOMMENDATION

6.1. CONCLUSION

The development of a web-based e-commerce application is a multifaceted process that requires careful planning, implementation, and testing. As businesses increasingly shift towards online platforms, the importance of creating a robust, user-friendly, and secure e-commerce system cannot be overstated. A well-designed GIHON e-commerce platform not only facilitates seamless transactions but also enhances customer engagement, optimizes inventory management, and supports marketing efforts.

Throughout the implementation and testing phases, it is essential to prioritize user experience, scalability, and security. By doing so, businesses can build trust with customers, ensuring that their sensitive information is protected while providing a smooth shopping experience. Moreover, the integration of analytics tools allows for data-driven decision-making, enabling businesses to adapt to changing market demands and customer preferences.

The successful deployment of an e-commerce application ultimately hinges on collaboration among stakeholders, continuous monitoring, and iterative improvements based on user feedback. As the online retail landscape evolves, businesses must remain agile and responsive to maintain a competitive edge.

6.2. RECOMMENDATIONS

To ensure the long-term success of a GIHON web-based e-commerce application, the following recommendations should be considered:

Focus on User Experience

Invest in UI/UX design to create an intuitive and visually appealing interface. Regularly gather user feedback to identify areas for improvement.

Implement features such as personalized recommendations and easy navigation to enhance customer satisfaction.

Enhance Security Measures

Employ robust security protocols, including SSL certificates, data encryption, and secure payment gateways, to protect user information.

Conduct regular security audits and vulnerability assessments to identify and mitigate risks proactively.

Optimize for Mobile

Ensure that the e-commerce platform is fully responsive and optimized for mobile devices. A significant portion of online shopping occurs on mobile, and a seamless mobile experience is crucial.

Consider developing a mobile app to provide an additional channel for customer engagement and convenience.

Implement Data Analytics

Utilize analytics tools to track customer behavior, sales trends, and inventory levels. This data can inform marketing strategies and inventory management.

Regularly analyze KPIs such as conversion rates, cart abandonment rates, and customer retention to assess performance and identify opportunities for improvement.

Plan for Scalability

Design the architecture of the e-commerce application to support scalability, allowing for increased traffic and transactions during peak seasons.

Consider cloud-based solutions that can scale resources up or down based on demand.

Integrate Third-Party Services

Leverage third-party services for payment processing, shipping, and customer relationship management to streamline operations and enhance functionality.

Ensure that integrations are secure and reliable to maintain a seamless user experience.

Continuous Improvement

Adopt an agile development approach that allows for iterative updates and enhancements based on user feedback and market trends.

Stay informed about emerging technologies and e-commerce trends to identify opportunities for innovation and improvement.

Provide Excellent Customer Support

Implement various customer support channels, including live chat, email, and phone support, to assist users effectively.

Create a comprehensive help center with FAQs and guides to empower users to find solutions independently.

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